# Improving the Performance of Cell-Edge Users in NOMA Systems using Cooperative Relaying

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***Abstract*—In this paper, we study performance improvement methods for a cell-edge user of two-user non-orthogonal multiple access (NOMA) systems in downlink scenarios. To this end, we propose two cooperative relaying schemes, namely on/off-full-duplex relaying (on/off-FDR) and on/off-half-duplex relaying (on/off-HDR) schemes. More specifically, in order to improve the performance of the cell-edge user, we consider a cell-center user as a relay, where either FDR or HDR can be employed to assist the direct NOMA transmission from a base station (BS) to the cell-edge user. An on/off mechanism is proposed to decide whether the cooperative relaying transmission is necessary or not. The on/off relaying decision is made based on the quality of the direct and relaying links from the BS to the cell-edge user. The performance of the two proposed schemes is investigated in terms of outage probability (OP) and sum throughput. Numerical results reveal that the proposed schemes not only provide essential outage performance improvements for the cell-edge user, but also are able to improve the sum throughput of the two-user NOMA systems. The advantages and drawbacks of each proposed scheme are highlighted and insightful discussions are provided.**

***Index Terms*—Non-orthogonal multiple access (NOMA), cooperative relaying, full-duplex relaying (FDR), half-duplex relaying (HDR), Markov chain, outage probability, sum throughput.**

## I. Introduction

Non-orthogonal multiple access (NOMA) has been considered as the latest member of the multiple access family, being a promising solution to improve spectral efficiency for the fifth generation (5G) of cellular communication systems [1]–[5]. For downlink NOMA, the underlying concept is to utilize the power domain for user multiplexing, which is different from conventional orthogonal multiple access (OMA) techniques (e.g., time/frequency/code division multiple access). For a preliminary explanation of the ideas behind NOMA, we consider a downlink scenario of a two-user NOMA system, in which a base station (BS) communicates concurrently with two users, one of which has a stronger channel condition (which is often located near the BS or at the cell-center) than the other (which is often a cell-edge user). At the BS, information signals of the two users are superposed with different power allocations, where the power allocation coefficient of the cell-edge user is higher than that of the cell-center user. At the receiver side of the cell-center user, a successive interference cancellation (SIC) receiver first decodes the message of the cell-edge user and then subtracts this message to decode the cell-center user's message. In contrast, at the cell-edge user, by utilizing the large power allocation different between the paired users, the message of the cell-edge user can be decoded directly since the message of the cell-center user can be considered as noise.

By using NOMA, the spectral efficiency can be significantly improved since both cell-center and cell-edge users are scheduled together and benefit from being assigned more bandwidth [2]. Also, NOMA can support more simultaneous connections since multiple users with different channel conditions are grouped to be served at the same time. As reported in [4], NOMA is able to improve cell-edge users with poor channel conditions, alleviating therefore the bottleneck issue of cell-edge throughput in cellular systems.

However, recent works in the literature have reported that the operation of NOMA raises a critical user throughput fairness issue between the cell-center and cell-edge users. Indeed, the achievable rate of the cell-edge users is often lower than that of the cell-center users [6]. In order to balance the data rate between cell-center and cell-edge users, the power allocation coefficient of the cell-center user has to be close to zero if the cell-edge user needs a data rate comparable to that of the cell-center user [7]. Hence, it may harm the quality-of-service (QoS) of the cell-center users since a major part of the power budget is allocated to cell-edge users, otherwise, it may compromise the reception reliability of the cell-edge users. Thus, the question is: how to solve the issue of fairness in performance between cell-center and cell-edge users, meanwhile improve the performance of cell-edge user?

The work of T. N. Do and B. An was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (Grant No. 2016R1D1A1B03934898) and by the Leading Human Resource Training Program of Regional Neo industry Through the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and future planning (Grant No. 2016H1D5A1910577). The work of D. B. da Costa was supported in part by CNPq under Grant 304301/2014-0. The work of T. Q. Duong was supported in part by the U.K. Royal Academy of Engineering Research Fellowship under Grant RF1415\14\22 and U.K. Engineering and Physical Sciences Research Council under Grant EP/P019374/1. This paper was presented in part at the IEEE Global Communications Conference (GLOBECOM), Singapore, Dec. 2017.

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### A. Related Works

One possible solution is to use cooperative relaying transmissions, where cell-center users serve as relays to help cell-edge user(s). This forms a new category, called cooperative NOMA systems. The authors in [8] proposed a cooperative NOMA transmission scheme, in which cell-center users exploit prior information available in NOMA system to help improve the reception reliability of cell-edge users. In [9], the authors analyzed the capacity of a two-user NOMA system, where a relay is a cell-center user. Considering best user selection among multiple available users, the authors in [10] considered a two-user NOMA network, where the best cell-center user is selected to serve as a half-duplex (HD) relay to help improve the outage performance of a selected cell-edge user. In [11], considering Nakagami-$m$ fading, the authors proposed a cooperative NOMA system, in which a dedicated amplify-and-forward relay assists a base station to communicate with multiple mobile users simultaneously. However, all of the aforementioned works only considered the cooperative transmission with HD operation.

It is noteworthy that in HD cooperative NOMA systems, the NOMA transmission is just performed in a half of a time block, and the remaining half time block is used for cooperative relaying. Consequently, the use of half-duplex relaying (HDR) induces spectral efficiency degradation. Thus, in order to overcome such problem, few literatures considered full-duplex relaying (FDR) as a solution. Recently, considering a two-user NOMA system, in which a cell-center user acts as a full-duplex (FD) relay since the direct link between a BS and a cell-edge user is not available, the authors in [12] carried out an optimal analysis to find the optimal value of the power allocation that minimizes the outage performance. Very recently, in [13], the authors investigated a cooperative NOMA system, in which a cell-center user works as a decode-and-forward (DF) FDR or HDR to help a cell-edge user assuming that a direct link between a BS and the cell-edge user may or may not exist. However, in these past works, the self-interference (SI) channel introduced by the FDR was not well characterized and investigated, i.e., the SI channel was simply assumed to follow Gaussian [12] or Rayleigh [13] distributions.

### B. Contributions

In this paper, we consider a downlink scenario of a two-user cooperative NOMA system, where either FDR or HDR can be employed at the cell-center user. Note that in such scenario, the cell-center not only processes its own data but also acts as a relay to assist the direct transmission from a BS to a cell-edge user. We raise an original question that has not been reported in the field of cooperative NOMA, i.e., when and for which conditions FDR or HDR should be employed? To answer this question, we propose an on/off relaying mechanism, which will decide whenever a cooperative relaying is necessarily needed for the cell-edge user. The idea of the on/off mechanism is inspired by the idea of the switch-and-stay combining technique [14]. The main contributions of the paper can be summarized as follows:

- We propose two cooperative relaying schemes, namely on/off-full-duplex relaying (on/off-FDR) and on/off-half-duplex relaying (on/off-HDR), aiming to enhance the performance of the cell-edge user in two-user NOMA systems. In particular, taking into account time varying characteristic of wireless channels, the proposed on/off mechanism will activate the relaying operation (FDR or HDR) employed at the cell-center user when the quality of direct link between the BS and the cell-edge is severely poor, i.e., falls below an on/off threshold, and vice-versa.
- Different from previous works, in this paper, the SI is assumed to follow a Rician $K$-factor distribution, which makes the proposed on/off-FDR more practical in addition to be more mathematically challenging. In order to analyze the performance of the two proposed schemes, we formulate the on/off mechanism as a two-state Markov chain model. We then carry out a rigorous and comprehensive outage performance analysis and derive closed-form expressions for the outage probability of the respective cell-center and cell-edge users. It noteworthy that the on/off cooperative NOMA systems using either FDR or HDR have their own mathematical performance analysis challenges, which have not been encountered in related works. Specifically, with the introduction of the on/off threshold, the outage performance now is more intricate to derive, since we have to considered the relative relationship between the multiple signal-to-noise ratio (SNR) thresholds. Consequently, one outage probability (OP) may have various closed-form expressions according to the different relative relationships of the SNR thresholds as presented in the appendixes of the manuscript.
- We show through representative numerical examples that the proposed schemes indeed improve the outage performance of the cell-edge user. However, the cell-center user suffers a performance loss due to the employment of relaying transmissions. In addition, the two proposed schemes increase the sum throughput of the cell-center and cell-edge users. Moreover, we achieve a throughput balance between the cell-center and cell-edge users, which alleviates the user throughput fairness issue.
- Besides, we also show that the use of either FDR, on/off-FDR, HDR, or on/off-HDR draw their own positive and negative effects on each user. In particular, employing FDR could degrade the performance of NOMA at moderate to high transmit SNR. Nevertheless, with the on/off mechanism, the proposed FDR scheme becomes more effective not only for the cell-edge user but also for the overall system performance. On the other hand, the on/off-HDR scheme generally performs better than the on/off-FDR scheme at high SNR and vice-versa. Additionally, the on/off-HDR scheme also eliminates the performance loss created by the HDR scheme without the on/off mechanism at the cell-center user.

**Notations**: $\mathbb{E}[\cdot]$ denotes expectation; $F_X(x)$, $\bar{F}_X(x)$, and $f_X(x)$ represent cumulative distribution function (CDF), complementary CDF (CCDF), and probability density function (PDF) of an arbitrary random variable $X$, respectively; $\Pr(\mathrm{A})$

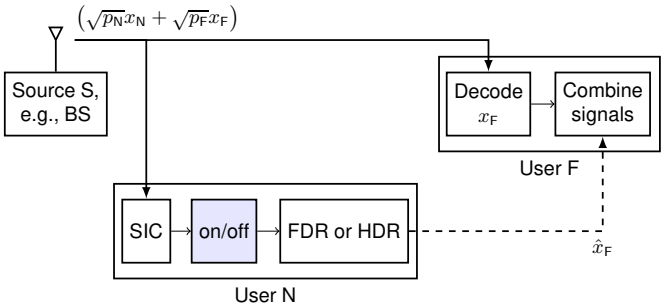



Fig. 1. Illustration of a two-user NOMA system with the on/off cooperative relaying mechanism.

stands for the probability that an event A occurs; $\mathcal{CN}(0, \sigma^2)$ denotes a circular symmetric complex Gaussian random variable with zero-mean and variance $\sigma^2$; $I_0(\cdot)$ designates the zeroth-order modified Bessel function of the first kind [15, Eq. (8.406.1)], $\mathrm{Ei}(\cdot)$ is the exponential integral function [15, Eq. (8.211.1)], $M_{\lambda,\mu}(\cdot)$ denotes the Whittaker function [15, Eq. (9.220.2)], $\gamma(\cdot,\cdot)$ denotes the incomplete gamma function [15, Eq. (8.350.1)].

## II. System Model

### A. System Description

Let us consider downlink transmission scenarios of two-user NOMA systems consisting of one source (e.g., a BS), one cell-center user, and one cell-edge user denoted by $\mathsf{S}$, User $\mathsf{N}$, and User $\mathsf{F}$, respectively, as depicted in Fig. 1. The source $\mathsf{S}$ and User $\mathsf{F}$ are HD single-antenna nodes while User $\mathsf{N}$ is a FD node equipped with two antennas.

Considering a channel from $\mathsf{X}$ to $\mathsf{Y}$, where $\mathsf{X} \in \{\mathsf{S}, \mathsf{N}\}, \mathsf{Y} \in \{\mathsf{N}, \mathsf{F}\}$, let $\tilde{h}_{\mathsf{XY}}$ and $G_{\mathsf{XY}}$ characterize the small-scale fading and large-scale path-loss effects of the $\mathsf{X} \rightarrow \mathsf{Y}$ channel. Assuming all wireless channels, except the SI channel, exhibit Rayleigh block flat fading. Since the small-scale fading magnitude $|\tilde{h}_{\mathsf{XY}}|$ follows Rayleigh distribution, the corresponding fading gain $|\tilde{h}_{\mathsf{XY}}|^2$ follows the exponential distribution, whose CDF, $F_{|\tilde{h}_{\mathsf{XY}}|^2}$, and PDF, $f_{|\tilde{h}_{\mathsf{XY}}|^2}$, can be expressed as $F_{|\tilde{h}_{\mathsf{XY}}|^2}(h) = 1 - e^{-\frac{h}{\tilde{\lambda}_{\mathsf{XY}}}}, f_{|\tilde{h}_{\mathsf{XY}}|^2}(h) = \frac{1}{\tilde{\lambda}_{\mathsf{XY}}} e^{-\frac{h}{\tilde{\lambda}_{\mathsf{XY}}}}$, respectively, where $\tilde{\lambda}_{\mathsf{XY}}$ denotes the mean of $|\tilde{h}_{\mathsf{XY}}|^2$. On the other hand, the large-scale path-loss can be modeled as $G_{\mathsf{XY}} = (d_{\mathsf{XY}}/d_0)^{-\epsilon}\mathcal{L}$, [16], where $d_{\mathsf{XY}}$ stands for the Euclidean between $\mathsf{X}$ and $\mathsf{Y}$, $d_0$ represents the reference distance, $\mathcal{L}$ means the measured power degradation at $d_0$ (in dB), and $\epsilon$ represents the path-loss exponent.

Thus, in this paper, a received signal at $\mathsf{Y}$ which is transmitted from $\mathsf{X}$, $y_{\mathsf{Y}}(t)$, can be expressed as

$$y_{\mathsf{Y}}(t) = \sqrt{P_{\mathsf{X}} G_{\mathsf{XY}}}\tilde{h}_{\mathsf{XY}} x(t) + n_{\mathsf{Y}}(t), \tag{1}$$

where $P_{\mathsf{X}}$ denotes the transmit power of $\mathsf{X}$, $x(t)$ represent the transmitted signal, and $n_{\mathsf{Y}}(t)$ characterize the additive white Gaussian noise (AWGN) at $\mathsf{Y}$.

In addition, for the sake of exposition, let $h_{\mathsf{XY}} \triangleq G_{\mathsf{XY}}\tilde{h}_{\mathsf{XY}}$. Consequently, the CDF, $F_{|h_{\mathsf{XY}}|^2}(h)$, and PDF, $f_{|h_{\mathsf{XY}}|^2}(h)$, of $|h_{\mathsf{XY}}|^2$ can be written as $F_{|h_{\mathsf{XY}}|^2}(h) = 1 - e^{-\frac{h}{\lambda_{\mathsf{XY}}}}, f_{|h_{\mathsf{XY}}|^2}(h) = \frac{1}{\lambda_{\mathsf{XY}}} e^{-\frac{h}{\lambda_{\mathsf{XY}}}}$, respectively, where $\lambda_{\mathsf{XY}} \triangleq G_{\mathsf{XY}}\tilde{\lambda}_{\mathsf{XY}} = \frac{\mathcal{L}}{(d_{\mathsf{XY}}/d_0)^{\epsilon}}\tilde{\lambda}_{\mathsf{XY}}$ denotes the mean of $|h_{\mathsf{XY}}|^2$. Note that $\lambda_{\mathsf{XY}}$ is obtained by relying on the fact that, for a non-negative continuous exponential random variable $Z \sim f_Z(z)$ and a real number $\alpha$, revoking [17, Theorem 5-1], it is shown that $\mathbb{E}[\alpha Z] = \alpha\mathbb{E}[Z]$. In addition, we further assume $|\tilde{h}_{\mathsf{XY}}|^2$ follows an exponential distribution with unit mean, as in [16].

### B. The Two-User NOMA System

*1) The NOMA Transmission at* $\mathsf{S}$*:* At the beginning of a time block, $\mathsf{S}$ broadcasts a superposed signal $\sqrt{p_{\mathsf{N}}}x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}}x_{\mathsf{F}}$, where $x_{\mathsf{N}}$ and $x_{\mathsf{F}}$ denote the normalized signals transmitted to Users $\mathsf{N}$ and $\mathsf{F}$, respectively, i.e., $\mathbb{E}[|x_{\mathsf{N}}|^2] = \mathbb{E}[|x_{\mathsf{F}}|^2] = 1$, and $p_{\mathsf{N}}$ and $p_{\mathsf{F}}$ denote the power allocation coefficients for Users $\mathsf{N}$ and $\mathsf{F}$, respectively. Following the principle of NOMA, we assume that $|h_{\mathsf{SN}}|^2 > |h_{\mathsf{SF}}|^2$, $p_{\mathsf{N}} < p_{\mathsf{F}}$, and $p_{\mathsf{N}} + p_{\mathsf{F}} = 1$ [1].

*2) Successive Interference Cancellation (SIC) at User* $\mathsf{N}$*:* The baseband signal for information decoding at User $\mathsf{N}$, $y_{\mathsf{SN}}$, can be expressed as

$$y_{\mathsf{SN}}^{\mathtt{nc}} = \sqrt{P_{\mathsf{S}}}h_{\mathsf{SN}}(\sqrt{p_{\mathsf{N}}}x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}}x_{\mathsf{F}}) + n_{\mathsf{N}}, \tag{2}$$

where $P_{\mathsf{S}}$ and $P_{\mathsf{N}}$ represent the transmit powers of $\mathsf{S}$ and User $\mathsf{N}$, respectively, and $n_{\mathsf{N}} \sim \mathcal{CN}(0, \sigma_{\mathsf{N}}^2)$.

The SIC receiver at User $\mathsf{N}$ first decodes $x_{\mathsf{F}}$ and removes this component from the received superposed signal; then, $x_{\mathsf{N}}$ is decoded. From (2), the signal-to-interference-plus-noise ratio (SINR) at User $\mathsf{N}$ to decode $x_{\mathsf{F}}$, $\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}}$, and the SNR at User $\mathsf{N}$ to decode $x_{\mathsf{N}}$, $\gamma_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{nc}}$, can be, respectively, expressed as

$$\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}} = \frac{P_{\mathsf{S}}p_{\mathsf{F}}|h_{\mathsf{SN}}|^2}{P_{\mathsf{S}}p_{\mathsf{N}}|h_{\mathsf{SN}}|^2 + \sigma_{\mathsf{N}}^2}, \tag{3}$$

$$\gamma_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{nc}} = \frac{P_{\mathsf{S}}p_{\mathsf{N}}|h_{\mathsf{SN}}|^2}{\sigma_{\mathsf{N}}^2}, \tag{4}$$

where the superscript "$\mathtt{nc}$" stands for "non-cooperative" mode. We further assume tht if User $N$ is able to decoded $x_{\mathsf{F}}$ and/or $x_{\mathsf{N}}$ in its SIC process, we assume that $x_{\mathsf{F}}$ and $x_{\mathsf{N}}$ are decoded correctly without any error.

*3) Information Decoding at User* $\mathsf{F}$*:* The received baseband observation at User $\mathsf{F}$ can be written as

$$y_{\mathsf{SF}}^{\mathtt{nc}} = \sqrt{P_{\mathsf{S}}}h_{\mathsf{SF}}(\sqrt{p_{\mathsf{N}}}x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}}x_{\mathsf{F}}) + n_{\mathsf{F}}, \tag{5}$$

where and $n_{\mathsf{F}} \sim \mathcal{CN}(0, \sigma_{\mathsf{F}}^2)$. In contrast with the cell-center user, the cell-edge user can directly decode its information signal since the cell-edge user is allocated with higher transmit power and thus the interference introduced by the information signal of the cell-center user can be considered as noise [1]. Thus, the received SNR at User $\mathsf{F}$ to decode $x_{\mathsf{F}}$ can be expressed as

$$\gamma_{\mathsf{SF}}^{\mathtt{nc}} = \frac{P_{\mathsf{S}}p_{\mathsf{F}}|h_{\mathsf{SF}}|^2}{P_{\mathsf{S}}p_{\mathsf{N}}|h_{\mathsf{SF}}|^2 + \sigma_{\mathsf{F}}^2}. \tag{6}$$

Next, we are going to describe four schemes: in Schemes I and III, we merely employ FDR and HDR in NOMA, respectively, while Schemes II and IV are the proposed on/off-HDR and on/off-HDR schemes, respectively. It is noteworthy that although the idea of Scheme I is simple, its performance

evaluation under our setting, i.e., Rician $K$-factor fading, has not been reported in the literature. In addition, the investigation of Schemes I and III aims to help readers easily follow the motivation and development of Schemes II and IV, respectively.

### C. The On/Off Mechanism

For either FDR or HDR strategies, let $R_{\mathsf{SF}}^{\mathsf{nc}}$ and $R_{\mathsf{SNF}}$ denote the data rate of the $\mathsf{S} \to \mathsf{F}$ link (in non-cooperative NOMA system) and $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ link (in cooperative NOMA system), respectively. The operation of the on/off mechanism can be explained as follows. Before data transmission, the BS uses channel state information (CSI), which is assumed to be perfectly estimated in this paper, to determine whether the cooperative relaying transmission should be used or not in the current time block based on an a predetermined on/off data rate threshold $R_0$ (bps/Hz). For illustration, suppose that the cooperative relaying transmission was inactivated (i.e., the BS communicates with User $\mathsf{F}$ on the $\mathsf{S} \to \mathsf{F}$ link) in the previous time block. Then, the BS compares the potential data rate of the $\mathsf{S} \to \mathsf{F}$ link with the on/off SNR threshold, there are two scenarios as follows. If $R_{\mathsf{SF}}^{\mathsf{nc}}$ is less than $R_0$, User $\mathsf{F}$ feedbacks to the BS that the cooperative relaying is needed and User $\mathsf{N}$ is announced to activate its relaying transmission, regardless of whether $R_{\mathsf{SNF}}$ is going to be greater or less than $R_0$. Otherwise, the BS communicates with User $\mathsf{F}$ only via the $\mathsf{S} \to \mathsf{F}$ link and the relaying transmission at User $\mathsf{N}$ remains inactivated.

We aim to design an on/off model that helps the transmission exploits the $\mathsf{S} \to \mathsf{F}$ link, as much opportunistic as it can, while activate the $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ relaying link only in case that the transmission actually needs. With such principle in mind, the on/off mechanism can be modeled as a two-state Markov chain with the "on" and "off" states being represented by "1" and "0", respectively, as depicted in Fig. 2. Specifically, in state 0, relaying transmission is *not* performed at User $\mathsf{N}$ (the BS communicates with User $\mathsf{F}$ only via the $\mathsf{S} \to \mathsf{F}$ link); in state 1, relaying transmission is performed at User $\mathsf{N}$ (the $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ relaying link exists, the BS communicates with User $\mathsf{F}$ using both $\mathsf{S} \to \mathsf{F}$ and $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ links). Consequently, there are four transition events between two states as follows:

Event 1: The off-off event: $R_{\mathsf{SF}}^{\mathsf{nc}} \geq R_0$,
Event 2: The off-on event: $R_{\mathsf{SF}}^{\mathsf{nc}} < R_0$,
Event 3: The on-off event: $R_{\mathsf{SNF}} < R_0$,
Event 4: The on-on event: $R_{\mathsf{SNF}} \geq R_0$.

From the transition events, it is noteworthy that: when $R_{\mathsf{SF}}^{\mathsf{nc}} < R_0$, the BS activates the relaying transmission performed at User $\mathsf{N}$, regardless of whether $R_{\mathsf{SNF}}$ is going to be greater or less than $R_0$; and when $R_{\mathsf{SNF}} < R_0$, the transmission turns into non-cooperative NOMA transmission, regardless of whether $R_{\mathsf{SF}}^{\mathsf{nc}}$ is going to be greater or less than $R_0$.

#### Benefit of The on/off Mechanism

The on/off mechanism not only helps the BS determine whether to use cooperative transmission to help the cell-edge user but also brings great advantages when it is integrated into HDR or FDR transmissions.

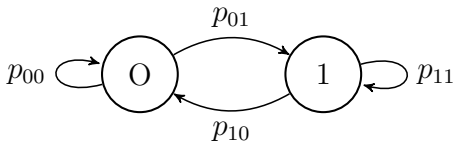



Fig. 2. The state transition diagram of the Markov chain model describing the on/off mechanism, where "1" and "0" represents for "on" and "off" decisions, respectively.

*For the HDR transmission:* It is known that in order to implement HDR, a time slot is divided into two equal sub-slots, where the first sub-slot is for direct transmission and the second one is for cooperative relaying. Such time division decreases the spectral efficiency in comparison to the conventional non-cooperative NOMA system.

It is noteworthy that the on/off-HDR scheme opportunistically avoids such spectral efficiency degradation. Specifically, when HDR is not employed at the cell-center user in a certain time slot, that time slot does not need to be divided, so that the NOMA transmission is implemented in the whole time slot. This feature of the on/off-HDR scheme is characterized via two kinds of SNR thresholds, e.g., for the case of the cell-center user, $\rho_1 = 2^{R_1} - 1$ indicates the SNR threshold when HDR is off and $\tilde{\rho}_1 = 2^{2R_1} - 1$ indicates the SNR threshold when HDR is on, where $R_1$ denotes the target data rate of the cell-center user. Similarly, for the case of the cell-edge user, $\rho_2 = 2^{R_2} - 1$ indicates the SNR threshold when HDR is off and $\tilde{\rho}_2 = 2^{2R_2} - 1$ indicates the SNR threshold when HDR is on, where $R_2$ denotes the target data rate of the cell-edge user.

*For the FDR transmission:* Since the use of HDR jeopardizes the spectral efficiency of NOMA schemes, another possible solution is to employ FDR. Specifically, by using a full-duplex cell-center user, the cooperative relaying can be implemented simultaneously with the direct NOMA transmission. As a result, the spectral efficiency of the conventional non-cooperative NOMA system is preserved. However, the use of FDR in NOMA system creates SI at the cell-center user and co-channel interference (CCI) at the cell-edge user since both direct and relaying transmission are performed at the same time on the same frequency.

The on/off-FDR scheme is able to occasionally avoid such interferences at the cell-center and cell-edge users. Particularly, in State "0" of the on/off-FDR scheme, the direct transmission is good enough to satisfy the Quality-of-Service of the cell-edge user, while the cell-edge and cell-center users are not suffered from SI and CCI, respectively.

It noteworthy that the proposed on/off mechanism can also be applied to NOMA systems, where multiple cell-center and cell-edge users communicating with a BS, the BS can select only one pair of cell-center and cell-edge users to serve in a certain time slot using a certain user selection criterion. Moreover, a pair of users can be randomly selected by the BS as in [18], or by following user selection schemes such as the best near best far (BNBF) selection scheme [10]. In order to deal with multi-user NOMA schemes, we may need advanced beamforming schemes, for instance, the multiple-user channel

state information based singular value decomposition (MU-CSI-SVD) beamforming scheme proposed in [19].

*Steady-state Probabilities*

Let $p_{00}, p_{01}, p_{10}$, and $p_{11}$ denote transition probabilities of events one to four, respectively. And let $\pi_1$ and $\pi_0$ denote the steady-state probabilities of the on and off status, respectively. The relationship between $\pi_1$ and $\pi_0$ associated with the described Markov chain can be written as [20]

$$\begin{aligned} \pi_0 &= p_{00}\pi_0 + p_{10}\pi_1, \\ \pi_1 &= p_{01}\pi_0 + p_{11}\pi_1, \\ 1 &= \pi_0 + \pi_1. \end{aligned} \tag{7}$$

Relying on the fact that $p_{00} = 1 - p_{01}$ and $p_{11} = 1 - p_{10}$, and after some manipulations, $\pi_0$ and $\pi_1$ can be, respectively, expressed as

$$\pi_0 = \frac{p_{10}}{1 - p_{00} + p_{10}} = \frac{p_{10}}{p_{01} + p_{10}}, \tag{8}$$

$$\pi_1 = \frac{p_{01}}{1 - p_{11} + p_{01}} = \frac{p_{01}}{p_{01} + p_{10}}. \tag{9}$$

### D. Scheme I: The FDR Scheme Without The on/off Mechanism

*1) Full-Duplex Relaying (FDR) at User* $\mathsf{N}$*:* In Scheme I, User $\mathsf{N}$ performs FDR to forward $x_{\mathsf{F}}$, which is decoded during its SIC operation, to User $\mathsf{F}$. Due to the effect of SI channel, the baseband signal for information decoding at User $\mathsf{N}$, $y_{\mathsf{SN}}$, can be expressed as

$$y_{\mathsf{SN}}^{\mathtt{FD}} = \sqrt{P_{\mathsf{S}}} h_{\mathsf{SN}} (\sqrt{p_{\mathsf{N}}} x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}} x_{\mathsf{F}}) + \sqrt{P_{\mathsf{N}}} h_{\mathsf{si}} \hat{x}_{\mathsf{F}} + n_{\mathsf{N}}, \tag{10}$$

where $\hat{x}_{\mathsf{F}}$ represents the normalized re-encoded signal forwarded by User $\mathsf{N}$ to User $\mathsf{F}$, $\mathbb{E}[|\hat{x}_{\mathsf{F}}|^2] = 1$, the processing delay at the User $\mathsf{N}$ for $\hat{x}_{\mathsf{F}}$ can be negligible; $h_{\mathsf{si}}$ stands for the SI channel incident on the receive antenna of User $\mathsf{N}$.

In the literature, the SI channel can be modeled as additive white Gaussian noise (AWGN), Rayleigh, or Rician distributed variables [21], [22]. In particular, the SI channel consists of two main components: the line-of-sight (LOS) component due to the direct link between the transmit and receive antennas, and the non-LOS component due to the signal reflections. Accordingly, the first tap of the SI channel can be modeled as a Rician distribution with large $K$-factor [16], [23]–[25] and the remaining channel taps are modeled as Rayleigh fading [25] or as Rician fading with smaller $K$-factor [24].

In our paper, we consider a simplified version of the SI channel, i.e., only the LOS component is significant, and the SI channel thus follows a $K$-factor Rician distribution, as done in [16]. Thus, the PDF, $f_{|h_{\mathsf{si}}|^2}(x)$, of the corresponding channel gain can be written as

$$f_{|h_{\mathsf{si}}|^2}(x) = \frac{(K+1)e^{-K}}{\lambda_{\mathsf{si}}} e^{-\frac{(K+1)x}{\lambda_{\mathsf{si}}}} I_0\left(2\sqrt{\frac{K(K+1)x}{\lambda_{\mathsf{si}}}}\right), \tag{11}$$

where $K$ denotes the Rician $K$-factor defined as the ratio of the powers of the light-of-sight (LOS) component to the scattered components, and $\lambda_{\mathsf{si}} = K/(K+1)$ [24] stands for the average fading power of the SI channel.

From (10), the SINR at User $\mathsf{N}$ to decode $x_{\mathsf{F}}$, $\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}}$, and the SINR to decode $x_{\mathsf{N}}$, $\gamma_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{FD}}$, can be, respectively, expressed as

$$\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}} = \frac{P_{\mathsf{S}} p_{\mathsf{F}} |h_{\mathsf{SN}}|^2}{P_{\mathsf{S}} p_{\mathsf{N}} |h_{\mathsf{SN}}|^2 + P_{\mathsf{N}} |h_{\mathsf{si}}|^2 + \sigma_{\mathsf{N}}^2}, \tag{12}$$

$$\gamma_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{FD}} = \frac{P_{\mathsf{S}} p_{\mathsf{N}} |h_{\mathsf{SN}}|^2}{P_{\mathsf{N}} |h_{\mathsf{si}}|^2 + \sigma_{\mathsf{N}}^2}, \tag{13}$$

where the superscript "$\mathtt{FD}$" indicates that FDR is performed at User $\mathsf{N}$.

*2) Receiving Observations at User* $\mathsf{F}$*:* In this scenario, i.e., the FDR is active, User $\mathsf{F}$ receives two incoming signals, one is the forwarded signal from User $\mathsf{N}$ and the other is the superposed signal transmitted by the source. In this case, the forwarded signal can be considered as the desired signal while the direct signal can be considered as interference, as in [26]. The baseband form of the observation at User $\mathsf{F}$ can be then expressed as

$$y_{\mathsf{NF}}^{\mathtt{FD}} = \underbrace{\sqrt{P_{\mathsf{N}}} h_{\mathsf{NF}} \hat{x}_{\mathsf{F}}}_{\text{desired component}} + \underbrace{\sqrt{P_{\mathsf{S}}} h_{\mathsf{SF}} (\sqrt{p_{\mathsf{N}}} x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}} x_{\mathsf{F}}) + n_{\mathsf{F}}}_{\text{interference + noise}}.$$

Thus, the received SNR of the relaying link at User $\mathsf{F}$ can be written as

$$\gamma_{\mathsf{NF}}^{\mathtt{FD}} = \frac{P_{\mathsf{N}} |h_{\mathsf{NF}}|^2}{P_{\mathsf{S}} |h_{\mathsf{SF}}|^2 + \sigma_{\mathsf{F}}^2}. \tag{14}$$

Considering the relaying link, i.e., the link $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$, the failure of one of the two hops leads to the failure of the whole relaying transmission. Thus, the end-to-end achievable SNR of the relaying channel can be expressed as

$$\gamma_{\mathsf{SNF}}^{\mathtt{FD}} = \min\{\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}}, \gamma_{\mathsf{NF}}^{\mathtt{FD}}\}. \tag{15}$$

### E. Scheme II: The on/off-FDR Scheme

Scheme II employs the on/off mechanism as described in subsection II-C where the FDR is adopted at User $\mathsf{N}$. It means that when the off-off and on-off events occur, the considered system turns into the same as the non-cooperative two-user NOMA system, in this case, the $\mathsf{N} \to \mathsf{F}$ channel does not exist. When the off-on and on-on events occur, FDR is activated, and Scheme II turns into the same as Scheme I.

The data rates associated to the $\mathsf{S} \to \mathsf{F}$ and $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ links can be expressed as

$$R_{\mathsf{SF}}^{\mathtt{nc}} = \log_2(1 + \gamma_{\mathsf{SF}}^{\mathtt{nc}}), \tag{16}$$

$$R_{\mathsf{SNF}}^{\mathtt{FD}} = \log_2(1 + \gamma_{\mathsf{SNF}}^{\mathtt{FD}}), \tag{17}$$

respectively. Thus, the transition probabilities used in Scheme II can be expressed as

$$p_{00} = \Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} \geq \rho_0), \tag{18}$$

$$p_{01} = \Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0), \tag{19}$$

$$p_{10} = \Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho_0), \tag{20}$$

$$p_{11} = \Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} \geq \rho_0), \tag{21}$$

where $\rho_0 \triangleq 2^{R_0} - 1$.

### F. Scheme III: The HDR Scheme Without The on/off Mechanism

We now turn our attention to HDR. In Scheme III, a cooperative transmission is carried out in two equal-time phases, namely broadcasting and relaying phases. In the broadcasting phase, S broadcasts the superposed signal using a non-cooperative two-user NOMA system to both Users N and F. In this case, the SNR/SINR at Users N and F, namely $\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}$, $\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}}$, and $\gamma^{\mathtt{HD}}_{\mathsf{SF}}$, can be expressed as the same as in (3), (4) and (6), respectively. In the relaying phase, since User N priorly knows $x_{\mathsf{F}}$ during its SIC process, User N forwards the re-encoded version $\hat{x}_{\mathsf{F}}$ to User F. The baseband form of the received signal at User F can be expressed as

$$y^{\mathtt{HD}}_{\mathsf{NF}} = \sqrt{P_{\mathsf{N}}} h_{\mathsf{NF}} \hat{x}_{\mathsf{F}} + n_{\mathsf{F}}. \tag{22}$$

Thus, the received SNR of the relaying link at User F can be written as

$$\gamma^{\mathtt{HD}}_{\mathsf{NF}} = \frac{P_{\mathsf{N}} |h_{\mathsf{NF}}|^2}{\sigma^2_{\mathsf{F}}}. \tag{23}$$

User F then combines the two signals received during the two phases using a maximal-ratio combining (MRC) technique. The end-to-end achievable SNR of the relaying channel can be expressed as

$$\gamma^{\mathtt{HD}}_{\mathsf{SNF}} = \min\{\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}, \gamma^{\mathtt{HD}}_{\mathsf{SF}} + \gamma^{\mathtt{HD}}_{\mathsf{NF}}\}. \tag{24}$$

### G. Scheme IV: The on/off-HDR Scheme

Scheme IV employs the on/off mechanism as described in subsection II-C where the HDR is adopted at User N. It means that when the off-off and on-off events occur, the considered system turns into the same as the non-cooperative two-user NOMA system, in this case, the $\mathsf{N} \to \mathsf{F}$ channel does not exist and the NOMA transmission is carried out in the full time slot. When the off-on and on-on events occur, HDR is activated, and Scheme IV turns into the same as Scheme III.

It is noteworthy that an on/off decision is made at the beginning of a time slot by comparing between two potential data rate, namely the data rate of the conventional non-cooperative $\mathsf{S} \to \mathsf{F}$ link and the data rate of the HDR $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ link. If HDR will be decided to be used, the slot will be equally divided, if not, the slot is preserved to be full. Therefore, the full-potential data rate of the $\mathsf{S} \to \mathsf{F}$ link is always $R^{\mathtt{nc}}_{\mathsf{SF}} = \log_2(1 + \gamma^{\mathtt{nc}}_{\mathsf{SF}})$. On the other hand, the data rate associated to the $\mathsf{S} \to \mathsf{N} \to \mathsf{F}$ link in Scheme IV can be expressed

$$R^{\mathtt{HD}}_{\mathsf{SNF}} = \frac{1}{2} \log_2(1 + \gamma^{\mathtt{HD}}_{\mathsf{SNF}}). \tag{25}$$

Thus, the transition probabilities used in Scheme IV can be expressed as

$$p_{00} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} \geq \rho_0), \tag{26}$$
$$p_{01} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} < \rho_0), \tag{27}$$
$$p_{10} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} < \tilde{\rho}_0), \tag{28}$$
$$p_{11} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} \geq \tilde{\rho}_0), \tag{29}$$

where $\tilde{\rho}_0 \triangleq 2^{2R_0} - 1$.

#### Remarks on the inter-node link

As can be observed from (21), the condition that the system decides to keep performing FDR at the cell-center user is $(\gamma^{\mathtt{FD}}_{\mathsf{SNF}} \geq \rho_0)$, which is equivalent to

$$\underbrace{(\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_0)}_{A_1} \text{ AND } \underbrace{(\gamma^{\mathtt{FD}}_{\mathsf{NF}} \geq \rho_0)}_{A_2},$$

where event $A_1$ means the cell-center user are able to successfully decode $x_{\mathsf{F}}$ during its SIC process, and event $A_2$ means the condition of the relaying link is qualified, i.e., its data rate is greater than or equal to the on/off data rate threshold, to help the cell-edge user.

As can be seen from (29), the condition that the system decides to keep performing HDF at the cell-center user is $(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} \geq \tilde{\rho}_0)$, which is equivalent to

$$\underbrace{(\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \tilde{\rho}_0)}_{B_1} \text{ AND } \underbrace{(\gamma^{\mathtt{HD}}_{\mathsf{NF}} + \gamma^{\mathtt{HD}}_{\mathsf{SF}} \geq \tilde{\rho}_0)}_{B_2},$$

where event $B_1$ means the cell-center user are able to successfully decode $x_{\mathsf{F}}$ during its SIC process, and event $B_2$ means the relaying is good enough so that the MRC of the SNRs of the direct link and the relaying link is greater than or equal to the on/off SNR threshold.

## III. Outage Performance Analysis

Recall that $R_1$ and $R_2$ (bps/Hz) denote the target data rates of Users N and F, respectively, $\rho_1 = 2^{R_1} - 1$ and $\rho_2 = 2^{R_2} - 1$ denote SNR thresholds of Users N and F, respectively, when a full time block is used, $\tilde{\rho}_1 = 2^{2R_1} - 1$ and $\tilde{\rho}_2 = 2^{2R_2} - 1$ denote SNR thresholds of Users N and F, respectively, when a half time block is used. Let $\bar{\gamma}_{\mathsf{S}} \triangleq \frac{P_{\mathsf{S}}}{\sigma^2_{\mathsf{S}}}$ and $\bar{\gamma}_{\mathsf{N}} \triangleq \frac{P_{\mathsf{N}}}{\sigma^2_{\mathsf{N}}}$ denote the transmit SNR at the BS and User N, respectively. In order to facilitate the performance analysis and for the sake of notational convenience, we assume that $\bar{\gamma}_{\mathsf{S}} = \bar{\gamma}_{\mathsf{N}} = \bar{\gamma}$, as in [13], [27]–[29]. For the sake of notational convenience, let $X \triangleq |h_{\mathsf{SN}}|^2$, $Y \triangleq |h_{\mathsf{NF}}|^2$, $Z \triangleq |h_{\mathsf{SF}}|^2$, and $T \triangleq |h_{\mathsf{si}}|^2$. Also, defined constants presented in Tables I and following self-defined functions (30)-(36) are going to be used along the developed analysis.

$$\Psi(\lambda, \rho) = 1 - e^{-\frac{\rho}{\lambda}} \tag{30}$$

$$\Omega(\rho) = \frac{1}{\lambda_{\mathsf{SF}}} e^{-\frac{\rho}{a_3 \lambda_{\mathsf{NF}}}} \left( \frac{\rho}{\lambda_{\mathsf{NF}}} + \frac{1}{\lambda_{\mathsf{SF}}} \right)^{-1} \tag{31}$$

$$\Xi(\kappa, \alpha) = \frac{\kappa(K+1)e^{-K}}{\lambda_{\mathsf{si}}} \frac{e^{\frac{K(K+1)}{2\lambda_{\mathsf{si}}} \left(\alpha + \frac{K+1}{\lambda_{\mathsf{si}}}\right)^{-1}}}{\sqrt{\left(\alpha + \frac{K+1}{\lambda_{\mathsf{si}}}\right) \frac{K(K+1)}{\lambda_{\mathsf{si}}}}} \times M_{-\frac{1}{2},0}\left( \frac{K(K+1)}{\lambda_{\mathsf{si}}} \left( \alpha + \frac{K+1}{\lambda_{\mathsf{si}}} \right)^{-1} \right) \tag{32}$$

$$\Lambda(\rho, \kappa, \alpha) = 1 - \Xi(\kappa, \alpha)\Omega(\rho) \tag{33}$$

TABLE I
SELF-DEFINED CONSTANTS

| | | | | |
|---|---|---|---|---|
| $a_1 = \bar{\gamma} p_{\mathsf{F}}$ | $\mu_0 = \frac{\rho_0}{a_1 - a_2\rho_0}$ | $\tilde{\mu}_0 = \frac{\tilde{\rho}_0}{a_1 - a_2\tilde{\rho}_0}$ | $\kappa_0 = e^{-\frac{\rho_0}{(a_1 - a_2\rho_0)\lambda_{\mathsf{SN}}}}$ | $\alpha_0 = \frac{a_3\rho_0}{(a_1 - a_2\rho_0)\lambda_{\mathsf{SN}}}$ |
| $a_2 = \bar{\gamma} p_{\mathsf{N}}$ | $\mu_1 = \frac{\rho_1}{a_2}$ | $\tilde{\mu}_1 = \frac{\tilde{\rho}_1}{a_2}$ | $\kappa_1 = e^{-\frac{\rho_1}{a_2\lambda_{\mathsf{SN}}}}$ | $\alpha_1 = \frac{a_3\rho_1}{a_2\lambda_{\mathsf{SN}}}$ |
| $a_3 = \bar{\gamma}$ | $\mu_2 = \frac{\rho_2}{a_1 - a_2\rho_2}$ | $\tilde{\mu}_2 = \frac{\tilde{\rho}_2}{a_1 - a_2\tilde{\rho}_2}$ | $\kappa_2 = e^{-\frac{\rho_2}{(a_1 - a_2\rho_2)\lambda_{\mathsf{SN}}}}$ | $\alpha_2 = \frac{a_3\rho_2}{(a_1 - a_2\rho_2)\lambda_{\mathsf{SN}}}$ |
| $\theta = p_{\mathsf{F}}/p_{\mathsf{N}}$ | $\nu_1 = (\mu_1 - \mu_0)\,\frac{a_3}{\mu_0}$ | $\nu_2 = (\mu_2 - \mu_0)\,\frac{a_3}{\mu_0}$ | $\kappa_3 = e^{-\frac{\max\{\rho_0,\rho_2\}}{(a_1 - a_2\max\{\rho_0,\rho_2\})\lambda_{\mathsf{SN}}}}$ | $\alpha_3 = \frac{a_3\max\{\rho_0,\rho_2\}}{(a_1 - a_2\max\{\rho_0,\rho_2\})\lambda_{\mathsf{SN}}}$ |

$$\Phi(\nu,\kappa,\alpha) = \sum_{l=0}^{\infty} \frac{\kappa(K+1)e^{-K}}{\lambda_{\mathsf{si}}(l!)^2}\left(\frac{K(K+1)}{\lambda_{\mathsf{si}}}\right)^{l} \times \left(\alpha + \frac{K+1}{\lambda_{\mathsf{si}}}\right)^{-(l+1)} \gamma\left(l+1, \left(\alpha + \frac{K+1}{\lambda_{\mathsf{si}}}\right)\nu\right) \tag{34}$$

$$\Theta(\mu) = \frac{e^{-\mu\chi}}{\chi} - \xi\Gamma(0,\mu\chi) + \sum_{q=2}^{\infty}\frac{(-1)^q\xi^q}{q!} \times \left[e^{-\mu\chi}\sum_{p=1}^{q-1}\frac{(p-1)!(-\chi)^{q-p-1}}{(q-1)!\mu^p} - \frac{(-\chi)^{q-1}}{(q-1)!}\mathrm{Ei}(-\mu\chi)\right] \tag{35}$$

$$\Upsilon(\mu,\rho) = 1 - e^{-\frac{\mu}{\lambda_{\mathsf{SF}}}} - \frac{1}{a_2a_3\lambda_{\mathsf{NF}}\lambda_{\mathsf{SF}}}e^{-\frac{\rho}{a_3\lambda_{\mathsf{NF}}} + \frac{a_1}{a_2a_3\lambda_{\mathsf{NF}}} + \frac{1}{a_2\lambda_{\mathsf{SF}}}} \times \left[\Theta(a_3\lambda_{\mathsf{NF}}) - \Theta\left(a_2a_3\lambda_{\mathsf{NF}}\mu + a_3\lambda_{\mathsf{NF}}\right)\right] \tag{36}$$

### *A. Outage Analysis of The Non-Cooperative Two-User NOMA Scheme*

The OP of a communication channel can be defined as the probability that the SNR of the channel falls below a predefined target data rate.

*1) User* $\mathsf{N}$*:* The OP of User $\mathsf{N}$ can be expressed as

$$\begin{aligned} P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}} &= \Pr(R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2) + \Pr(R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1) \\ &= \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2) + \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1), \end{aligned} \tag{37}$$

where $R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} = \log_2(1+\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}})$, $R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} = \log_2(1+\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}})$, $\rho_1 = 2^{R_1} - 1$ and $\rho_2 = 2^{R_2} - 1$.

*Proposition 1:* A closed-form expression for $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}}$ can be written as

$$P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}} = \begin{cases} \Psi(\lambda_{\mathsf{SN}},\mu_1), & \text{if } \rho_2 < \theta, \forall\rho_1, \mu_1 > \mu_2, \\ \Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } \rho_2 < \theta, \forall\rho_1, \mu_1 \leq \mu_2, \end{cases} \tag{38}$$

and $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}} = 1$ if $\rho_2 \geq \theta, \forall\rho_1$.

*Proof:* We can re-express $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}}$ in (37) as

$$P^{\mathtt{nc}}_{\mathrm{out},\mathsf{N}} = \Pr\left(\frac{a_1X}{a_2X+1} < \rho_2\right) + \Pr\left(\frac{a_1X}{a_2X+1} \geq \rho_2, a_2X < \rho_1\right). \tag{39}$$

The first probability in the right-hand side of (39) can be rewritten as $\Pr\left((a_1 - a_2\rho_2)X < \rho_2\right)$. Note that, this probability is always equal to 1 if $a_1 - a_2\rho_2 \leq 0$. Relying on this fact, and after some manipulations, the result in Proposition 1 can be attained. ■

*2) User* $\mathsf{F}$*:* Outage event occur at User $\mathsf{F}$ if $x_{\mathsf{F}}$ cannot be decoded at User $\mathsf{F}$. Thus,

$$\begin{aligned} P^{\mathtt{nc}}_{\mathrm{out},\mathsf{F}} &= \Pr(R^{\mathtt{nc}}_{\mathsf{SF}} < R_2) \\ &= \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} < \rho_2), \end{aligned} \tag{40}$$

and with some algebraic manipulations, $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{F}}$ can be attained as presented in the following proposition, whose proof is omitted.

*Proposition 2:* A closed-form expression for $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{F}}$ can be written as

$$P^{\mathtt{nc}}_{\mathrm{out},\mathsf{F}} = \Psi(\lambda_{\mathsf{SF}},\mu_2), \tag{41}$$

if $\rho_2 < \theta$, otherwise, $P^{\mathtt{nc}}_{\mathrm{out},\mathsf{F}} = 1$.

### *B. Outage Analysis of Scheme I*

*1) User* $\mathsf{N}$*:* Outage events occur at User $\mathsf{N}$ in Scheme I when either $x_{\mathsf{F}}$ or $x_{\mathsf{N}}$ cannot be decoded at User $\mathsf{N}$, which can be mathematically expressed as

$$\begin{aligned} P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}} &= \Pr(R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2) + \Pr(R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1), \\ &= \Pr\left(\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2\right) + \Pr\left(\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1\right), \end{aligned} \tag{42}$$

where $R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} = \log_2(1+\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}})$, $R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}} = \log_2(1+\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}})$.

*Proposition 3:* A closed-form expression for $P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}}$ can be expressed as

$$P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}} = \begin{cases} 1 - \Xi(\kappa_1,\alpha_1), & \text{if } \rho_2 < \theta, \mu_1 > \mu_2, \forall\rho_1, \\ 1 - \Xi(\kappa_2,\alpha_2), & \text{if } \rho_2 < \theta, \mu_1 \leq \mu_2, \forall\rho_1, \end{cases} \tag{43}$$

and $P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}} = 1$ if if $\rho_2 \geq \theta, \forall\rho_1$.

*Proof:* From (42), $P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}}$ can be rewritten as

$$P^{\mathtt{sI}}_{\mathrm{out},\mathsf{N}} = \Pr\left(\frac{a_1X}{a_2X+a_3T+1} < \rho_2\right) + \Pr\left(\frac{a_1X}{a_2X+a_3T+1} \geq \rho_2, \frac{a_2X}{a_3T+1} < \rho_1\right). \tag{44}$$

*Lemma 1:* The following probability can be derived as $\Pr\left((a_1X)/(a_2X+a_3T+1) < \rho\right) = 1 - \Xi(\kappa,\alpha)$ if $\rho < \theta$, otherwise, it is equal to 1.

*Proof:* With the help of [15, Eq. (6.614.3)], and after some algebraic manipulation, the result in Lemma 1 can be obtained. ■

By invoking Lemma 1, and relying on the fact that $\frac{a_3\rho_2 t+\rho_2}{a_1-a_2\rho_2} < \frac{a_3\rho_1 t+\rho_1}{a_2}, \forall t \in [0,\infty)$ if $\frac{\rho_2}{a_1-a_2\rho_2} < \frac{\rho_1}{a_2}$, (43) can be attained. ■

*2) User* F*:* Outage events occur at User F in Scheme I when the end-to-end SINR $\gamma_{\mathsf{SNF}}^{\mathtt{FD}}$ falls below the SNR threshold $\rho_2$, which can be mathematically expressed as

$$\begin{aligned} P_{\text{out},\mathsf{F}}^{\mathtt{sI}} &= \Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} < R_2) \\ &= \Pr\left(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho_2\right). \end{aligned} \tag{45}$$

*Proposition 4:* A closed-form expression for $P_{\text{out},\mathsf{F}}^{\mathtt{sI}}$ can be derived as

$$P_{\text{out},\mathsf{F}}^{\mathtt{sI}} = 1 - \Xi(\kappa_2,\alpha_2)\Omega(\rho_2), \tag{46}$$

if $\rho_2 < \theta, \forall \rho_1$, otherwise, $P_{\text{out},\mathsf{F}}^{\mathtt{sI}} = 1$.

*Proof:* $P_{\text{out},\mathsf{F}}^{\mathtt{sI}}$ can be rewritten as

$$P_{\text{out},\mathsf{F}}^{\mathtt{sI}} = \Pr\left(\min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_2\right). \tag{47}$$

By combining the result in Lemma 1 with our following result, i.e., $\int_0^\infty F_Y\left(\frac{a_3\rho_2 z+\rho_2}{a_3}\right) f_Z(z)dz = 1 - \Omega(\rho_2)$, we attain $P_{\text{out},\mathsf{F}}^{\mathtt{sI}}$ as presented in (46). ■

### C. Outage Analysis of Scheme II

From (8), (9), $\pi_0$ and $\pi_1$ in Scheme II can be, respectively, obtained as

$$\pi_0 = \frac{\Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho_0)}{\Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0) + \Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho_0)}, \tag{48}$$

$$\pi_1 = \frac{\Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0)}{\Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0) + \Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho_0)}. \tag{49}$$

By invoking Lemma 1, and after some manipulations, the probability $\Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho)$ can be obtained as

$$\begin{aligned} \Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho) &= \Pr\left(\min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho\right) \\ &= \underbrace{1 - \Xi(\kappa,\alpha)\Omega(\rho)}_{\triangleq \Lambda(\rho,\kappa,\alpha)}, \end{aligned} \tag{50}$$

if $\rho < \theta$, otherwise, $\Pr(\gamma_{\mathsf{SNF}}^{\mathtt{FD}} < \rho) = 1$. Thus, $\pi_0$ and $\pi_1$ can be obtained as

$$\pi_0 = \frac{\Lambda(\rho_0,\kappa_0,\alpha_0)}{\Psi(\lambda_{\mathsf{SF}},\mu_0) + \Lambda(\rho_0,\kappa_0,\alpha_0)}, \tag{51}$$

$$\pi_1 = \frac{\Psi(\lambda_{\mathsf{SF}},\mu_0)}{\Psi(\lambda_{\mathsf{SF}},\mu_0) + \Lambda(\rho_0,\kappa_0,\alpha_0)}, \tag{52}$$

if $\rho_0 < \theta$, otherwise, $\pi_0 = \pi_1 = 1/2$.

*1) User* N*:* The definition of the OP of User N can be formulated as follows. On one hand, if the system is in State 0, there are two events may happens, namely the off-off and off-on events, if the system is in Sate 1, there are two events may happens, namely the on-off and on-on events. On the other hand, regardless of the status of the FDR function, namely on or off, outage events occur at User N due to two reasons, namely: i) the SIC receiver of User N cannot decode $x_{\mathsf{F}}$, or ii) User N successfully decodes $x_{\mathsf{F}}$ but cannot decode its own message $x_{\mathsf{N}}$.

Relying on these facts, and by making use of the total probability theory [17], the OP of User N, $P_{\text{out},\mathsf{N}}^{\mathtt{nc}}$ can be written as

$$\begin{aligned} P_{\text{out},\mathsf{N}}^{\mathtt{sII}} = \pi_1\big[&\Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} \geq R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}} < R_2) \\ &+ \Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} \geq R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}} \geq R_2, R_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{FD}} < R_1) \\ &+ \Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}} < R_2) \\ &+ \Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}} \geq R_2, R_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{nc}} < R_1)\big] \\ &+ \pi_0\big[\Pr(R_{\mathsf{SF}}^{\mathtt{nc}} \geq R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}} < R_2) \\ &+ \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} \geq R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{nc}} \geq R_2, R_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{nc}} < R_1) \\ &+ \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}} < R_2) \\ &+ \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{FD}} \geq R_2, R_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{FD}} < R_1)\big]. \end{aligned} \tag{53}$$

*Proposition 5:* A closed-form expression for $P_{\text{out},\mathsf{N}}^{\mathtt{sII}}$ can be expressed as

$$P_{\text{out},\mathsf{N}}^{\mathtt{sII}} = \pi_1 \sum_{i=1}^{4} N_{2,i} + \pi_0 \sum_{i=5}^{8} N_{2,i}, \tag{54}$$

where $N_{2,i}$'s, $i = 1,\ldots,8$, are presented in (72), (74), (79), (84), (86), (88), (90), and (92), respectively.

*Proof:* See Appendix A. ■

*2) User* F*:* Relying on the total probability theorem [17], the OP of User F in Scheme II can be expressed as

$$\begin{aligned} P_{\text{out},\mathsf{F}}^{\mathtt{sII}} = \pi_0\big[&\Pr(R_{\mathsf{SF}}^{\mathtt{nc}} \geq R_0, R_{\mathsf{SF}}^{\mathtt{nc}} < R_2) \\ &+ \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} < R_0, R_{\mathsf{SNF}}^{\mathtt{FD}} < R_2)\big] \\ &+ \pi_1\big[\Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} \geq R_0, R_{\mathsf{SNF}}^{\mathtt{FD}} < R_2) \\ &+ \Pr(R_{\mathsf{SNF}}^{\mathtt{FD}} < R_0, R_{\mathsf{SF}}^{\mathtt{nc}} < R_2)\big]. \end{aligned} \tag{55}$$

*Proposition 6:* A closed-form expression for $P_{\text{out},\mathsf{F}}^{\mathtt{sII}}$ can be attained as

$$P_{\text{out},\mathsf{F}}^{\mathtt{sII}} = \pi_0(F_{2,1} + F_{2,2}) + \pi_1(F_{2,3} + F_{2,4}), \tag{56}$$

where $F_{2,i}$'s, $i = 1,\ldots,4$ are presented in (94), (97), (99), and (101), respectively.

*Proof:* See Appendix B. ■

### D. Outage Analysis of Scheme III

*1) User* N*:* The OP of User N in Scheme III can be expressed as

$$\begin{aligned} P_{\text{out},\mathsf{N}}^{\mathtt{sIII}} = &\Pr\left(R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < R_2\right) \\ &+ \Pr\left(R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} \geq R_2, R_{\mathsf{SN},x_{\mathsf{N}}}^{\mathtt{HD}} < R_1\right), \end{aligned} \tag{57}$$

where

$$R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}=\frac{1}{2}\log_2(1+\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}), R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}}=\frac{1}{2}\log_2(1+\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}}).$$

*Proposition 7:* A closed-form expression for $P^{\mathtt{sIII}}_{\text{out},\mathsf{N}}$ can be derived as

$$P^{\mathtt{sIII}}_{\text{out},\mathsf{N}}=\begin{cases}\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_1), & \text{if } \tilde{\rho}_2<\theta,\forall\tilde{\rho}_1,\tilde{\mu}_1>\tilde{\mu}_2,\\ \Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2), & \text{if } \tilde{\rho}_2<\theta,\forall\tilde{\rho}_1,\tilde{\mu}_1\leq\tilde{\mu}_2,\end{cases} \tag{58}$$

and $P^{\mathtt{sIII}}_{\text{out},\mathsf{N}}=1$, if $\tilde{\rho}_2\geq\theta,\forall\tilde{\rho}_1$.

*Proof:* $P^{\mathtt{sIII}}_{\text{out},\mathsf{N}}$ can be rewritten as

$$\begin{aligned}P^{\mathtt{sIII}}_{\text{out},\mathsf{N}}&=\Pr\left(\frac{a_1X}{a_2X+1}<\tilde{\rho}_2\right)\\&\quad+\Pr\left(\frac{a_1X}{a_2X+1}\geq\tilde{\rho}_2,a_2X<\tilde{\rho}_1\right).\end{aligned} \tag{59}$$

By meticulously considering the relative relationship between $\tilde{\rho}_1$, $\tilde{\rho}_2$, and $\theta$, and after some algebraic steps, (58) can be attained. ■

*2) User* $\mathsf{F}$*:* Supposing that $x_{\mathsf{F}}$ is successfully decoded at User $\mathsf{N}$, the data rate achieved using MRC at User $\mathsf{F}$ can be expressed as

$$R^{\mathtt{mrc}}_{\mathsf{F}}=\frac{1}{2}\log_2(1+\gamma^{\mathtt{HD}}_{\mathsf{SF}}+\gamma^{\mathtt{HD}}_{\mathsf{NF}}). \tag{60}$$

The OP of User $\mathsf{F}$ in Scheme III can be expressed as

$$\begin{aligned}P^{\mathtt{sIII}}_{\text{out},\mathsf{F}}&=\Pr\left(R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2,R^{\mathtt{HD}}_{\mathsf{SF}}<R_2\right)\\&\quad+\Pr\left(R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{mrc}}_{\mathsf{F}}<R_2\right),\end{aligned} \tag{61}$$

where $R^{\mathtt{HD}}_{\mathsf{SF}}=\frac{1}{2}\log_2(1+\gamma^{\mathtt{HD}}_{\mathsf{SF}})$.

*Proposition 8:* A closed-form expression for $P^{\mathtt{sIII}}_{\text{out},\mathsf{F}}$ can be achieved as

$$\begin{aligned}P^{\mathtt{sIII}}_{\text{out},\mathsf{F}}&=\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2)\Psi(\lambda_{\mathsf{SF}},\tilde{\mu}_2)\\&\quad+[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2)]\,\Upsilon(\tilde{\mu}_2,\tilde{\rho}_2),\end{aligned} \tag{62}$$

if $\tilde{\rho}_2<\theta$, otherwise, $P^{\mathtt{sIII}}_{\text{out},\mathsf{F}}=1$.

*Proof:* See Appendix C. ■

### *E. Outage Analysis of Scheme IV*

By invoking Lemma 3, and after some manipulations, the steady-state probabilities $\pi_0$ and $\pi_1$ of the Markov model used in Scheme IV can be written as

$$\begin{aligned}\pi_0&=\frac{\Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}}<\tilde{\rho}_0)}{\Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}}<\tilde{\rho}_0)+\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}\geq\rho_0)}\\&=\frac{1-[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)][1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)]}{1-[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)][1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)]+\Psi(\lambda_{\mathsf{SF}},\mu_0)},\end{aligned} \tag{63}$$

$$\begin{aligned}\pi_1&=\frac{\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_0)}{\Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}}<\tilde{\rho}_0)+\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}\geq\rho_0)}\\&=\frac{\Psi(\lambda_{\mathsf{SF}},\mu_0)}{1-[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)][1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)]+\Psi(\lambda_{\mathsf{SF}},\mu_0)},\end{aligned} \tag{64}$$

respectively, if $\rho_0<\tilde{\rho}_0<\theta$, and

$$\pi_0=\frac{1}{1+\Psi(\lambda_{\mathsf{SF}},\mu_0)},\quad \pi_1=\frac{\Psi(\lambda_{\mathsf{SF}},\mu_0)}{1+\Psi(\lambda_{\mathsf{SF}},\mu_0)}, \tag{65}$$

if $\rho_0<\theta,\tilde{\rho}_0\geq\theta$, and $\pi_0=\pi_1=1/2$ if $\theta\leq\rho_0<\tilde{\rho}_0$.

*1) User* $\mathsf{N}$*:* The OP of User $\mathsf{N}$ in Scheme IV can be expressed as

$$\begin{aligned}P^{\mathtt{sIV}}_{\text{out},\mathsf{N}}=\pi_1\big[&\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}\geq R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)\\&+\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}\geq R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)\\&+\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}<R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)\\&+\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}<R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)\big]\\+\pi_0\big[&\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)\\&+\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)\\&+\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)\\&+\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)\big].\end{aligned} \tag{66}$$

*Proposition 9:* A closed-form expression for $P^{\mathtt{sIV}}_{\text{out},\mathsf{N}}$ can be derived as

$$P^{\mathtt{sIV}}_{\text{out},\mathsf{N}}=\pi_1\sum_{i=1}^{4}N_{4,i}+\pi_0\sum_{i=5}^{8}N_{4,i}, \tag{67}$$

where $N_{4,i}$'s, $i=1,\ldots,8$, are presented in (110), (112), (114), (116), (118), (120), (122), and (124), respectively.

*Proof:* See Appendix D. ■

*2) User* $\mathsf{F}$*:* The OP of User $\mathsf{F}$ in Scheme IV can be written as

$$\begin{aligned}P^{\mathtt{sIV}}_{\text{out},\mathsf{F}}=\pi_0\big[&\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SF}}<R_2)\\&+\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2,R^{\mathtt{HD}}_{\mathsf{SF}}<R_2)\\&+\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{mrc}}_{\mathsf{F}}<R_2)\big]\\+\pi_1\big[&\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}\geq R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2,R^{\mathtt{HD}}_{\mathsf{SF}}<R_2)\\&+\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}\geq R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{mrc}}_{\mathsf{F}}<R_2)\\&+\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}<R_0,R^{\mathtt{nc}}_{\mathsf{SF}}<R_2)\big].\end{aligned} \tag{68}$$

*Proposition 10:* A closed-form expression for $P^{\mathtt{sIV}}_{\text{out},\mathsf{F}}$ can be attained as

$$P^{\mathtt{sIV}}_{\text{out},\mathsf{F}}=\pi_0\sum_{i=1}^{3}F_{4,i}+\pi_1\sum_{i=4}^{6}F_{4,i} \tag{69}$$

where $F_{4,i}$'s, $i=1,\ldots,6$ are presented in (126), (128), (131), (133), (135), and (137), respectively.

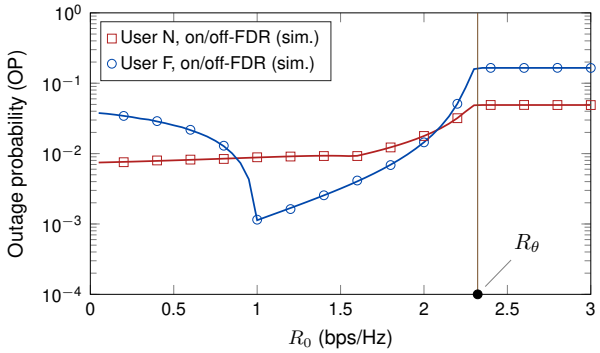
*Proof:* See Appendix E. ■

## IV. Numerical Results

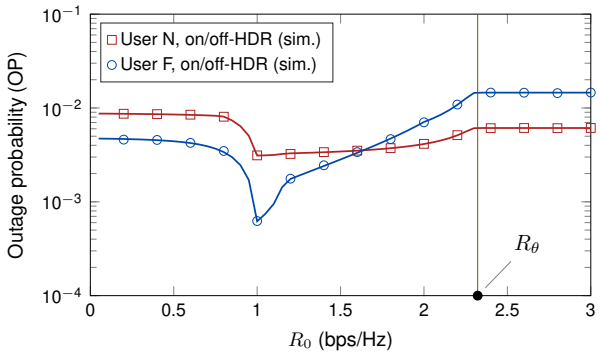
In this section, we present representative numerical results to illustrate the achievable performance of the proposed schemes. It is evident that (see Figs. 3, 5) the simulation results well matched with the analytical results which confirms the correctness of our developed analysis.[1]

In our plots, unless otherwise stated, we set that the distance between $\mathsf{S}$ and User $\mathsf{F}$ is $d_{\mathsf{SF}}=10$ m, the distance between $\mathsf{S}$ and User $\mathsf{N}$ is $d_{\mathsf{SN}}=3$ m, the distance between User $\mathsf{N}$ and

[1]Please note that the infinite series representations involved in $\Phi(\nu,\kappa,\alpha)$ and $\Theta(\mu)$ can be truncated without sacrificing numerical accuracy when number of terms used is high enough, e.g., 50 terms in each series representation was used to generate the analytical results in this section. Also, it is worthwhile to say that computational efficiency of the derived expressions is high, with the curves being plotted almost instantaneously.

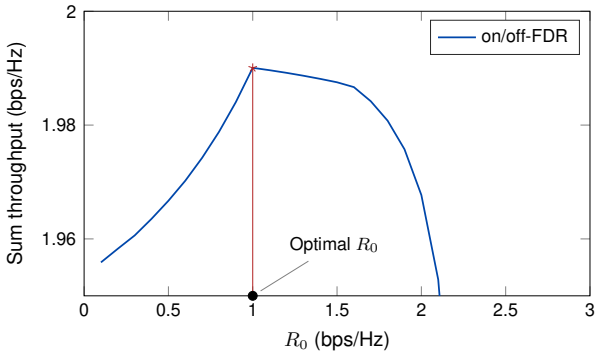



(a) Scheme II: the on/off-FDR scheme

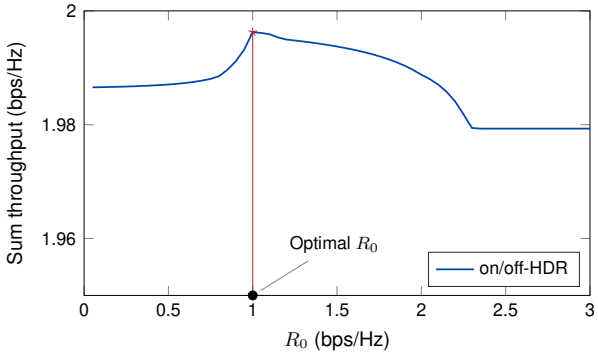



(b) Scheme IV: the on/off-HDR scheme

Fig. 3. OPs of Users N and F as a function of $R_0$ (bps/Hz) with the transmit SNR = 45 (dBm), where $R_\theta = \log_2(\theta + 1)$. Solid lines are theoretical results.




(a) Scheme II: the on/off-FDR scheme




(b) Scheme IV: the on/off-HDR scheme

Fig. 4. Illustrations of optimal values of $R_0$ that maximizes the sum throughput of Schemes II and IV with the transmit SNR of 45 (dBm).

User F is $d_{\mathsf{NF}} = d_{\mathsf{SF}} - d_{\mathsf{SN}}$,[2] the reference distance $d_0 = 1$ m, and the power degradation at $d_0$ is $\mathcal{L} = 30$ (dB), the path-loss exponent $\epsilon = 2.7$; the $K$-factor of Rician fading $K = 25$ (dB) for typical omni-directional antennas [25], fixed target data rates of Users N and F, $R_1 = R_2 = 1$ (bps/Hz). Recall that Schemes II and IV are referred to the proposed on/off-FDR and on/off-HDR schemes, respectively, Schemes I and III are referred to the FDR and HDR schemes without the on/of mechanism, respectively.

Please note that, in the comparison with the dynamic power allocation mechanism, the fixed mechanism has been widely used since it does not increase the complexity of the performance analysis while still reflects the principle as well as the performance efficiency of NOMA. And adopting a dynamic power allocation mechanism is indeed out of the scope of this paper, since this paper aims to evaluating the essential performance improvement of the proposed on/off scheme for cooperative NOMA systems. Thus, In the this paper, we adopt fixed power allocation coefficients, i.e., $p_{\mathsf{F}} = 0.8, p_{\mathsf{N}} = 1 - p_{\mathsf{F}} = 0.2$, which is the same setting in some closely relevant works, i.e., [10], [13], [18].

In Figs. 3 and 4, we examine the effect of the on/off threshold, $R_0$ (bps/Hz), on the performance of the proposed schemes in terms of outage probability and sum throughput, respectively. From this examination, we are going to point out the optimal value of $R_0$ that maximizes the sum throughput of the system in each scheme.

As can be seen from Fig. 3, OPs of Users N and F in both Schemes II and IV significantly vary as $R_0 < R_\theta$, where $R_\theta = \log_2(\theta + 1)$, and become steady as $R_0 \geq R_\theta$. One possible reason is that, by looking at the steady state probabilities, when $R_0 \geq R_\theta$, which equivalent to $\rho_0 \geq \theta$, the system will not transit from one state to the other but keeps staying in a certain state ("on" or "off" is depended on its past state). On the other hand, as shown in Fig. 3, by choosing an appropriate value of $R_0$, User F is able to achieve better outage performance than User N does, which is one of the main purpose of the proposed schemes, i.e., balancing the performance difference between the cell-center and cell-edge users.

Next, we are going to investigate the sum throughput of Schemes II and IV in order to numerically point out the optimal value of $R_0$ in each scheme. In particular, considering delay-sensitive communications, the sum throughput $\mathcal{T}^{\mathtt{sch}}$, where $\mathtt{sch} \in \{\mathtt{nc}, \mathtt{sI}, \mathtt{sII}, \mathtt{sIII}, \mathtt{sIV}\}$, can be defined as [31]

$$\mathcal{T}^{\mathtt{sch}} = (1 - P^{\mathtt{sch}}_{\mathrm{out},\mathsf{N}})R_1 + (1 - P^{\mathtt{sch}}_{\mathrm{out},\mathsf{F}})R_2. \tag{70}$$

Please note that the effect of transmission time on throughput is already considered in $\rho_i$ and $\tilde{\rho}_i$, $i = 1, 2$. In Figs. 4a and 4b, we plot the sum throughput of Schemes II and IV as a function of $R_0$, respectively. As can be observed, we obtain the maximum $\mathcal{T}^{\mathtt{sII}}$ at $R_0 = 1$ (bps/Hz) and the maximum $\mathcal{T}^{\mathtt{sIV}}$ at $R_0 = 1$ (bps/Hz), which are numerically found using MATLAB. According to our study, this remark of the optimal

[2]Please note that a line network formed by the base station S, User N, and User F is well assumed in the literature [10], [13], [30].

TABLE II
THROUGHPUTS (BPS/HZ) OF USER N AND USER F, AND SUM THROUGHPUT AT 30 (DBM)

| | Non-coop | only-FDR | on/off-FDR | only-HDR | on/off-HDR |
|---|---|---|---|---|---|
| Throughput of User N | 0.906 | 0.824 | 0.849 | 0.747 | 0.865 |
| Throughput of User F | 0.434 | 0.560 | 0.818 | 0.510 | 0.695 |
| Sum throughput | 1.340 | 1.384 | 1.667 | 1.257 | 1.560 |

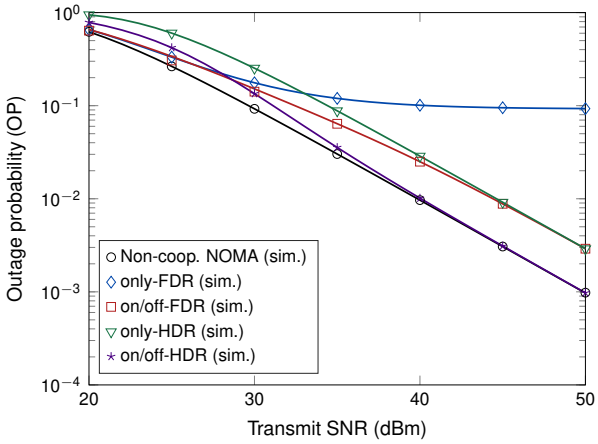
$R_0$'s is held for all values of SNR considered in this section. Also, as displayed in Fig. 3, at these optimal $R_0$'s, User F in Scheme II obtains a minimum OP, while both Users N and F in Scheme IV obtain minimum OPs. In the sequel of this the paper, we use this optimal on/off threshold, i.e., $R_0 = 1$ (bps/Hz) for both Scheme II and IV, respectively, to fairly evaluate the achievable performance of the proposed schemes.

In Figs. 5a and 5b, we plot the OP of Users N and F, respectively, in Schemes I-IV and the non-cooperative NOMA scheme as a function of the transmit SNR, $\bar{\gamma}$ (dBm). Regarding User F, as shown in Fig. 5b, both proposed schemes are able to lower the OP of User F in comparison to the cooperative NOMA systems without on/off mechanism or the non-cooperative NOMA, reflecting the purpose of the proposed schemes. On the other hand, Scheme II outperforms Scheme IV at low transmit SNR regime, i.e., $\bar{\gamma} \leq 40$ (dBm), and vice-versa in the high SNR regime, i.e., $\bar{\gamma} > 40$ (dBm). Regarding User N, as can be seen, the OP of User N increases when the relaying schemes are employed, which means User N suffers a performance loss in comparison to that of the non-cooperative NOMA. Nevertheless, different relaying schemes result in different performance losses. For example, with the on/off-HDR scheme, the outage performance of User N is least jeopardized in comparison with the other schemes.

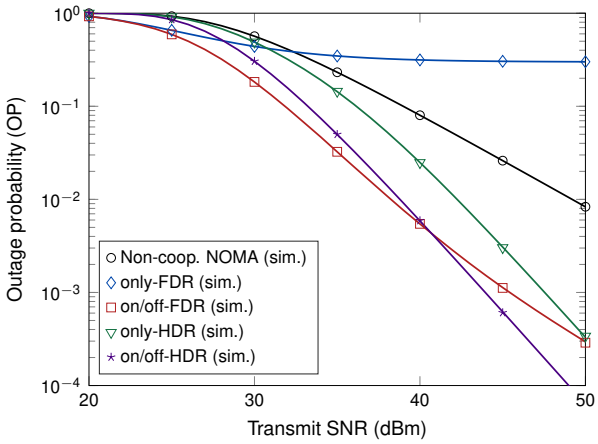
Solely speaking on FDR, the use FDR is beneficial when the transmit SNR is low, since the SI at User N is not severe. However, the affect of SI is stronger as $\bar{\gamma}$ increases, and the performance of FDR become worse. Nevertheless, with the on/off mechanism, we are able to avoid the effect of SI occasionally. For example, when $\mathsf{S} \rightarrow \mathsf{N}$ link is weak, the relaying link is thus activated. Now considering HDR, User N does not suffer any SI but the time for NOMA transmission is now only a half of a time block. With the on/off mechanism, we can occasionally avoid such transmission time loss.

In general, at both Users N and F, the use of the cooperative relaying on/off mechanism in FDR or HDR clearly provides better performance in comparison with the FDR or HDR schemes without on/off mechanism. Additionally, the benefit of the user of the on/off mechanism is more significant on FDR than on HDR.

Next, we investigate the performance improvement of the proposed schemes in terms of the sum throughput using analytical results presented in Fig. 6. As can be seen, in comparison with the non-cooperative NOMA scheme or the cooperative NOMA with out the on/off mechanism, both Schemes II and IV are able to significantly improve the sum throughput, excepts for the case of Scheme IV at quite low SNR, i.e., $\bar{\gamma} < 28$ (dBm). A possible reason for this result is




(a) User N




(b) User F

Fig. 5. OPs of a) User N and b) User F, respectively, as a function of SNR (dBm). Solid lines are theoretical results.

that by using the proposed schemes, the throughput of the cell-edge user can be significantly improved while the throughput of the cell-center just slightly jeopardizes.

As shown in Table. II, at the given transmit SNR of 30 (dBm), in the conventional non-cooperative NOMA, $\mathcal{T}_{\mathsf{N}}^{\mathtt{nc}} = 0.906$ while $\mathcal{T}_{\mathsf{F}}^{\mathtt{nc}} = 0.434$, which shows a big throughput gap between cell-center and cell-edge users, and the sum throughput is $\mathcal{T}^{\mathtt{nc}} = 1.340$; by using the on/off-HDR scheme, we achieve the sum throughput is $\mathcal{T}^{\mathtt{sIV}} = 1.560$, in which $\mathcal{T}_{\mathsf{N}}^{\mathtt{sIV}} = 0.865$ and $\mathcal{T}_{\mathsf{F}}^{\mathtt{sIV}} = 0.695$; and by using the on/off-FDR we achieve even higher sum throughput, i.e., $\mathcal{T}^{\mathtt{sIV}} = 1.667$, in which $\mathcal{T}_{\mathsf{N}}^{\mathtt{sIV}} = 0.849$ and $\mathcal{T}_{\mathsf{F}}^{\mathtt{sIV}} = 0.818$. Thus, it is conceivable that the fairness issue in throughput between the cell-center and cell-edge user is resolved by using the two proposed schemes.

We are going to further investigate the self-interference introduced by full-duplex transmission considering two kinds of antennas used in FDR at User N, namely omni-directional and directional antennas. From [25], by using omni-directional, the self-interference channel Rician factor is approximately 25 dB, while using directional antennas could decrease the self-interference channel Rician factor to 0 dB. We plot the OP of Users N and F in Schemes I and II under two kinds of antennas

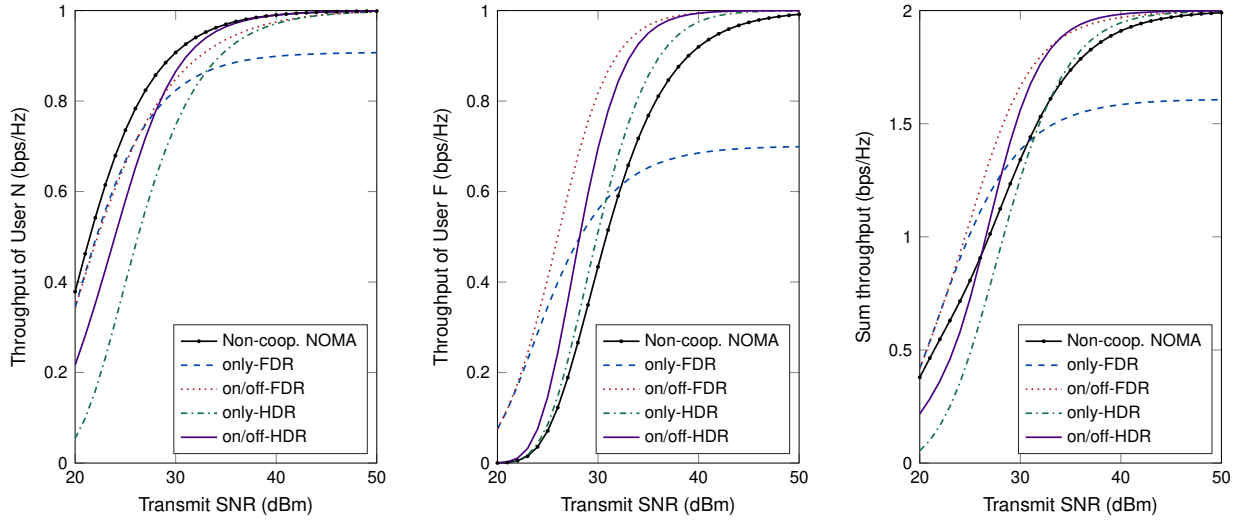
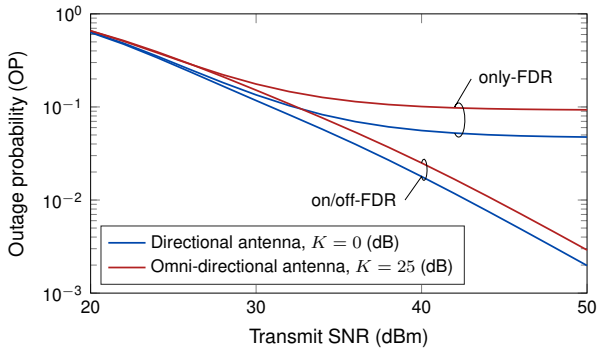
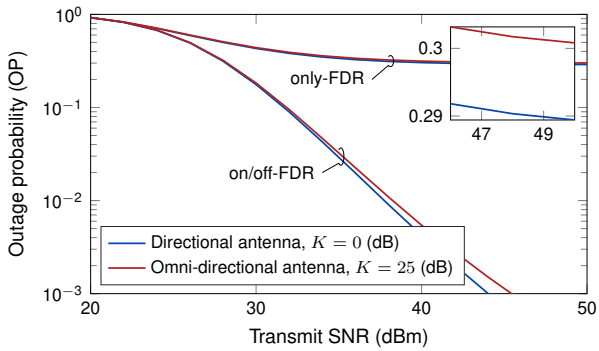



Fig. 6. Throughputs of User N, User F, and the sum throughput (bps/Hz), respectively, as a function for SNR (dBm).




(a) User N




(b) User F

Fig. 7. Illustration of the effect of the SI channel, i.e., $K$-factor on the outage performance of Users N and F in Schemes I and II.

in Figs. 7a and 7b, respectively. As can be observed in Fig. 7, the more the self-interference can be canceled, i.e., the $K$ factor is smaller, the lower OP we can achieve at both users. In addition, the self-interference suppression has strong affect on the outage performance of User N on the considered SNR range, while this effect is weaker on the outage performance of User F and occurs in some specific SNR values. This can be explained by looking at the SINR of Users N and F. For example, in the case of Scheme II, the SI involves in the whole SIC process of $x_{\mathsf{F}}$ and $x_{\mathsf{N}}$ at User N, while the SI just partly involves in the message decoding of user F, i.e., the $x_{\mathsf{F}}$ on the relaying link.

## V. Conclusions

In this paper, we have studied performance improvements for a cell-edge user in a two-user NOMA system. In particular, we have proposed two cooperative relaying schemes, namely the on/off-FDR and on/off-HDR schemes, which have been employed at the cell-center user to opportunistically assist the cell-edge user. Closed-form expressions for the OP of both cell-center and cell-edge users have been derived. The numerical results have showed that the use of cooperative relaying indeed improves the performance of the cell-edge user but also introduces a performance loss at the cell-center user. The two proposed schemes outperform either the cooperative FDR and HDR NOMA without the on/off mechanism or the conventional non-cooperative NOMA under some certain settings. On the other hand, the on/off-FDR scheme achieves better performance than the on/off-HDR scheme at low transmit SNR regime and vice-versa at high transmit SNR regime. Additionally, the on/off-FDR and on/off-HDR scheme are able to reduce the performance gap between the cell-center and cell-edge users, which suitably alleviate the user throughput fairness issue in NOMA.

## Appendix A
## Proof of Proposition 5

Let $N_{2,1} = \Pr(R^{\mathsf{FD}}_{\mathsf{SNF}} \geq R_0, R^{\mathsf{FD}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$\begin{aligned} N_{2,1} &= \Pr(\gamma^{\mathsf{FD}}_{\mathsf{SNF}} \geq \rho_0, \gamma^{\mathsf{FD}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2) \\ &= \Pr\left(\frac{a_1 X}{a_2 X + a_3 T + 1} \geq \rho_0, \right. \\ &\qquad \left. \frac{a_3 Y}{a_3 Z + 1} \geq \rho_0, \frac{a_1 X}{a_2 X + a_3 T + 1} < \rho_2\right). \end{aligned} \tag{71}$$

By considering the relative relationship between $\rho_0, \rho_2$, and $\theta$, and after some manipulations, $N_{2,1}$ can be obtained as

$$N_{2,1} = \begin{cases} \Xi(\kappa_0, \alpha_0)\Omega(\rho_0), & \text{if } \rho_0 < \theta, \rho_2 \geq \theta, \\ \big[\Xi(\kappa_0, \alpha_0) & \\ \quad -\Xi(\kappa_2, \alpha_2)\big]\Omega(\rho_0), & \text{if } \rho_0 < \theta, \rho_2 < \theta, \rho_0 < \rho_2, \end{cases} \tag{72}$$

and $N_{2,1} = 0$, if $(\rho_0 < \theta, \rho_2 < \theta, \rho_0 \geq \rho_2)$ or $\rho_0 \geq \theta$.

Let $N_{2,2} = \Pr(R^{\mathtt{FD}}_{\mathsf{SNF}} \geq R_0, R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$\begin{aligned} N_{2,2} &= \Pr(\gamma^{\mathtt{FD}}_{\mathsf{SNF}} \geq \rho_0, \gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1) \\ &= \Pr\left(\frac{a_3 Y}{a_3 Z + 1} \geq \rho_0\right) \Pr\left(\frac{a_1 X}{a_2 X + a_3 T + 1}\right. \\ &\quad \left. \geq \max\{\rho_0, \rho_2\}, \frac{a_2 X}{a_3 T + 1} < \rho_1\right). \end{aligned} \tag{73}$$

By considering the relative relationship between $\max\{\rho_0, \rho_2\}$, $\alpha_1$, $\alpha_3$, and $\theta$, and after some manipulations, $N_{2,2}$ can be obtained as

$$N_{2,2} = \Omega(\rho_0)[\Xi(\kappa_3, \alpha_3) - \Xi(\kappa_1, \alpha_1)], \tag{74}$$

if $\max\{\rho_0, \rho_2\} < \theta, \mu_1 > \max\{\rho_0, \rho_2\}$, otherwise, $N_{2,2} = 0$.

Let $N_{2,3} = \Pr(R^{\mathtt{FD}}_{\mathsf{SNF}} < R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$\begin{aligned} N_{2,3} &= \Pr(\gamma^{\mathtt{FD}}_{\mathsf{SNF}} < \rho_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2) \\ &= \Pr\left(\min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_0,\right. \\ &\quad \left. \frac{a_1 X}{a_2 X + 1} < \rho_2\right). \end{aligned} \tag{75}$$

For the case $\rho_2 < \theta$, after some algebraic steps, $N_{2,3}$ can be rewritten as

$$\begin{aligned} N_{2,3} &= \int_0^\infty \int_0^{\mu_2} f_X(x) f_T(t) dx dt \\ &\quad - \Pr\left(\frac{a_3 Y}{a_3 Z + 1} \geq \rho_0\right) \\ &\quad \times \int_0^\infty \int_0^{\mu_2} \Pr\left(\frac{a_1 x}{a_2 x + a_3 t + 1} \geq \rho_0\right) f_X(x) f_T(t) dx dt \\ &= \Psi(\lambda_{\mathsf{SN}}, \mu_2) - \Omega(\rho_0) N_{2,3,C}, \end{aligned} \tag{76}$$

where $N_{2,3,C} = \int_0^\infty \int_0^{\mu_2} \Pr\left(\frac{a_1 x}{a_2 x + a_3 t + 1} \geq \rho_0\right) f_X(x) f_T(t) dx dt$.

*Lemma 2:* A following integral can be obtained as follows:

$$\begin{aligned} &\int_0^\nu \kappa \frac{(K+1)e^{-K}}{\lambda_{\mathsf{si}}} e^{-\left(\alpha + \frac{K+1}{\lambda_{\mathsf{si}}}\right)t} I_0\left(2\sqrt{\frac{K(K+1)}{\lambda_{\mathsf{si}}} t}\right) dt \\ &= \Phi(\nu, \kappa, \alpha). \end{aligned} \tag{77}$$

*Proof:* Relying on the series representation of $I_0(\cdot)$ function [15, Eq. (8.447.1)], then making use of [15, Eq. 3.381.1], $\Phi(\nu, \kappa, \alpha)$ can be obtained. ■

As can be observed, $N_{2,3,C} = 0$ if $\rho_0 \geq \theta$. For the case $\rho_0 < \theta$, since $\frac{a_1 x}{a_2 x + a_3 t + 1} \geq \rho_0$ is equivalent to $x \geq \frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0}$. Thus, by invoking Lemma 2, $N_{2,3,C}$ can be derived as

$$\begin{aligned} N_{2,3,C} &= \int_0^{\nu_2} \int_{\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0}}^{\frac{\rho_2}{a_1 - a_2 \rho_2}} f_X(x) f_T(t) dx dt \\ &= \Phi(\nu_2, \kappa_0, \alpha_0) - \Phi(\nu_2, \kappa_2, 0), \end{aligned} \tag{78}$$

if $\mu_0 < \mu_2$, otherwise, $N_{2,3,C} = 0$. By combining the above steps, $N_{2,3}$ can be obtained as in (79), and $N_{2,3} = 1$ if $\rho_0 \geq \theta, \rho_2 \geq \theta$.

Let $N_{2,4} = \Pr(R^{\mathtt{FD}}_{\mathsf{SNF}} < R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$\begin{aligned} N_{2,4} &= \Pr(\gamma^{\mathtt{FD}}_{\mathsf{SNF}} < \rho_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1) \\ &= \Pr\left(\min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_0,\right. \\ &\quad \left. \frac{a_1 X}{a_2 X + 1} \geq \rho_2, a_2 X < \rho_1\right). \end{aligned} \tag{80}$$

It can be observed that $N_{2,4} = 0$ if $\rho_2 \geq \theta$. For the case $\rho_2 < \theta$, after some manipulations, $N_{2,4}$ can be expressed as

$$\begin{aligned} N_{2,4} &= \int_0^\infty \int_{\mu_2}^{\mu_1} \Pr\left(\min\left\{\frac{a_1 x}{a_2 x + a_3 t + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_0\right) \\ &\quad \times f_X(x) f_T(t) dx dt \\ &= \Psi(\lambda_{\mathsf{SN}}, \mu_1) - \Psi(\lambda_{\mathsf{SN}}, \mu_2) - \Omega(\rho_0) N_{2,4,C}, \end{aligned} \tag{81}$$

if $\mu_1 > \mu_2$, otherwise, $N_{2,4} = 0$, where $N_{2,4,C} = \int_0^\infty \int_{\mu_2}^{\mu_1} \Pr\left(\frac{a_1 x}{a_2 x + a_3 t + 1} \geq \rho_0\right) f_X(x) f_T(t) dx dt$. As can be observed, if $\rho_0 \geq \theta$, $N_{2,4,C} = 0$. For the case $\rho_0 < \theta$, we can see that if $\frac{\rho_0}{a_1 - a_2 \rho_0} > \frac{\rho_2}{a_1 - a_2 \rho_2}$, then $\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0} > \mu_2, \forall t \in [0, \infty)$. Next, the further consider is that $\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0} < \mu_1$, which is equivalent to $t < \nu_1$, where $\nu_1 = \left(\frac{\rho_1}{a_2} - \frac{\rho_0}{a_1 - a_2 \rho_0}\right) \frac{a_1 - a_2 \rho_0}{a_3 \rho_0}$. If $\frac{\rho_0}{a_1 - a_2 \rho_0} \geq \frac{\rho_1}{a_2}$, $N_{2,4,C} = 0$, otherwise, $N_{2,4,C}$ can be obtained as

$$\begin{aligned} N_{2,4,C} &= \int_0^{\nu_1} e^{-\frac{a_3 \rho_0 t + \rho_0}{(a_1 - a_2 \rho_0)\lambda_X}} f_T(t) dt - \int_0^{\nu_1} e^{\frac{\rho_1}{a_2 \lambda_X}} f_T(t) dt \\ &= \Phi(\nu_1, \kappa_0, \alpha_0) - \Phi(\nu_1, \kappa_1, 0), \end{aligned} \tag{82}$$

For the case $\frac{\rho_0}{a_1 - a_2 \rho_0} \leq \frac{\rho_2}{a_1 - a_2 \rho_2}$, if $\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0} < \mu_2$ then $0 \leq t \leq \nu_2$, where $\nu_2 = \left(\frac{\rho_2}{a_1 - a_2 \rho_2} - \frac{\rho_0}{a_1 - a_2 \rho_0}\right) \frac{a_1 - a_2 \rho_0}{a_3 \rho_0}$, otherwise, if $\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0} \geq \mu_2$ then $\nu_2 \leq t \leq \nu_1$. Note that $\nu_2 < \nu_1$ since, at this step, $\frac{\rho_1}{a_2} > \frac{\rho_2}{a_1 - a_2 \rho_2}$. Thus, $N_{2,4,C}$ can be obtained as

$$\begin{aligned} N_{2,4,C} &= \int_0^{\nu_2} \int_{\mu_2}^{\mu_1} f_X(x) f_T(t) dx dt \\ &\quad + \int_{\nu_2}^{\nu_1} \int_{\frac{a_3 \rho_0 t + \rho_0}{a_1 - a_2 \rho_0}}^{\mu_1} f_X(x) f_T(t) dx dt \\ &= \Phi(\nu_2, \kappa_2, 0) + \Phi(\nu_1, \kappa_0, \alpha_0) \\ &\quad - \Phi(\nu_1, \kappa_1, 0) - \Phi(\nu_2, \kappa_0, \alpha_0). \end{aligned} \tag{83}$$

After combining the above steps, $N_{2,4}$ can be obtained as in (84), otherwise, $N_{2,4} = 0$.

$$N_{2,3}=\begin{cases}\Psi(\lambda_{\mathsf{SN}},\mu_2)-\Omega(\rho_0)\left[\Phi(\nu_2,\kappa_0,\alpha_0)-\Phi(\nu_2,\kappa_2,0)\right], & \text{if } \rho_0<\theta,\rho_2<\theta,\mu_0<\mu_2,\\ \Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } (\rho_0<\theta,\rho_2<\theta,\mu_0\geq\mu_2) \text{ or } (\rho_0\geq\theta,\rho_2<\theta),\\ \Lambda(\rho_0,\kappa_0,\alpha_0), & \text{if } \rho_0<\theta,\rho_2\geq\theta.\end{cases} \tag{79}$$

$$N_{2,4}=\begin{cases}\Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2)-\Omega(\rho_0)\big[\Phi(\nu_2,\kappa_2,0) & \\ \quad+\Phi(\nu_1,\kappa_0,\alpha_0)-\Phi(\nu_1,\kappa_1,0)-\Phi(\nu_2,\kappa_0,\alpha_0)\big], & \text{if } \rho_0<\theta,\rho_2<\theta,\mu_0\geq\mu_2,\\ \Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2)-\Omega(\rho_0)\big[\Phi(\nu_1,\kappa_0,\alpha_0)-\Phi(\nu_1,\kappa_1,0)\big], & \text{if } \rho_0<\theta,\rho_2<\theta,\mu_2<\mu_0<\mu_1,\\ \Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } (\rho_0<\theta,\rho_2<\theta,\mu_2<\mu_0,\mu_0\geq\mu_1)\\ & \quad\text{or } (\rho_0\geq\theta,\rho_2<\theta,\mu_1>\mu_2).\end{cases} \tag{84}$$

Let $N_{2,5}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)$, which can be rewritten as

$$\begin{aligned}N_{2,5}&=\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}\geq\rho_0,\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}<\rho_2)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}\geq\rho_0,\frac{a_1X}{a_2X+1}<\rho_2\right),\end{aligned} \tag{85}$$

after some manipulations, $N_{2,5}$ can be obtained as

$$N_{2,5}=\begin{cases}\big[1-\Psi(\lambda_{\mathsf{SF}},\mu_0)\big]\Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } \rho_0<\theta,\rho_2<\theta\\ 1-\Psi(\lambda_{\mathsf{SF}},\mu_0), & \text{if } \rho_0<\theta,\rho_2\geq\theta,\end{cases} \tag{86}$$

and $N_{2,5}=0$ if $\rho_0\geq\theta,\forall\rho_2$.

Let $N_{2,6}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)$, which can be rewritten as

$$\begin{aligned}N_{2,6}&=\Pr\left(\gamma^{\mathtt{nc}}_{\mathsf{SF}}\geq\rho_0,\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}}\geq\rho_2,\gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}}<\rho_1\right)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}\geq\rho_0,\frac{a_1X}{a_2X+1}\geq\rho_2,a_2X<\rho_1\right),\end{aligned} \tag{87}$$

and after some manipulations, $N_{2,6}$ can be obtained as

$$N_{2,6}=[1-\Psi(\lambda_{\mathsf{SF}},\mu_0)][\Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2)], \tag{88}$$

if $\rho_0<\theta,\rho_2<\theta,\mu_1>\mu_2$, otherwise, $N_{2,6}=0$.

Let $N_{2,7}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}}<R_2)$, which can be rewritten as

$$\begin{aligned}N_{2,7}&=\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_0,\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}}<\rho_2)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}<\rho_0,\frac{a_1X}{a_2X+a_3T+1}<\rho_2\right),\end{aligned} \tag{89}$$

and after some manipulations, $N_{2,7}$ can be obtained as

$$N_{2,7}=\begin{cases}\Psi(\lambda_{\mathsf{SF}},\mu_0)\big[1-\Xi(\kappa_2,\alpha_2)\big], & \text{if } \rho_0<\theta,\rho_2\geq\theta,\\ 1-\Xi(\kappa_2,\alpha_2), & \text{if } \rho_0<\theta,\rho_2\geq\theta,\\ 1-\Xi(\kappa_2,\alpha_2), & \text{if } \rho_0\geq\theta,\rho_2<\theta,\end{cases} \tag{90}$$

and $N_{2,7}=1$ if $\rho_0\geq\theta,\rho_2\geq\theta$.

Let $N_{2,8}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}}<R_1)$, which can be rewritten as

$$\begin{aligned}N_{2,8}&=\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_0,\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq\rho_2,\gamma^{\mathtt{FD}}_{\mathsf{SN},x_{\mathsf{N}}}<\rho_1)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}<\rho_0,\frac{a_1X}{a_2X+a_3T+1}\geq\rho_2,\right.\\&\qquad\left.\frac{a_2X}{a_3T+1}<\rho_1\right),\end{aligned} \tag{91}$$

and after some manipulations, $N_{2,8}$ can be obtained as

$$N_{2,8}=\begin{cases}\Psi(\lambda_{\mathsf{SF}},\mu_0)\big[\Xi(\kappa_2,\alpha_2) & \\ \quad-\Xi(\kappa_1,\alpha_1)\big], & \text{if } \rho_0<\theta,\rho_2<\theta,\mu_1<\mu_2,\\ \Xi(\kappa_2,\alpha_2)-\Xi(\kappa_1,\alpha_1), & \text{if } \rho_0\geq\theta,\rho_2<\theta,\mu_1<\mu_2,\end{cases} \tag{92}$$

otherwise, $N_{2,8}=0$.

## Appendix B
## Proof of Proposition 5

Let $F_{2,1}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}\geq R_0,R^{\mathtt{nc}}_{\mathsf{SF}}<R_2)$, which can be rewritten as

$$\begin{aligned}F_{2,1}&=\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}\geq\rho_0,\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_2)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}\geq\rho_0,\frac{a_1Z}{a_2Z+1}<\rho_2\right),\end{aligned} \tag{93}$$

and after some algebraic steps, $F_{2,1}$ can be obtained as

$$F_{2,1}=\begin{cases}\Psi(\lambda_{\mathsf{SF}},\mu_2) & \\ \quad-\Psi(\lambda_{\mathsf{SF}},\mu_0), & \text{if } \rho_0<\theta,\rho_2<\theta,\rho_0<\rho_2,\\ \Psi(\lambda_{\mathsf{SF}},\mu_0), & \text{if } \rho_0<\theta,\rho_2\geq\theta,\end{cases}$$

and $F_{2,1}=0$ if $(\rho_0<\theta,\rho_2<\theta,\rho_0\geq\rho_2)$ or $\rho_0\geq\theta$.

Let $F_{2,2}=\Pr(R^{\mathtt{nc}}_{\mathsf{SF}}<R_0,R^{\mathtt{FD}}_{\mathsf{SNF}}<R_2)$, which can be rewritten as

$$\begin{aligned}F_{2,2}&=\Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_0,\gamma^{\mathtt{FD}}_{\mathsf{SNF}}<\rho_2)\\&=\Pr\left(\frac{a_1Z}{a_2Z+1}<\rho_0,\right.\\&\qquad\left.\min\left\{\frac{a_1X}{a_2X+a_3T+1},\frac{a_3Y}{a_3Z+1}\right\}<\rho_2\right).\end{aligned} \tag{94}$$

For the case $\rho_0\geq\theta$, $F_{2,2}$ can be obtained as $F_{2,2}=\Pr\left(\min\left\{\frac{a_1X}{a_2X+a_3T+1},\frac{a_3Y}{a_3Z+1}\right\}<\rho_2\right)=\Lambda(\rho_2,\kappa_2,\alpha_2)$, if

$\rho_2 < \theta$, otherwise, $F_{2,2} = 1$. For the case $\rho_0 < \theta$, by conditioning on $T = t$ first, and then $Z = z$, and after some manipulations, $F_{2,2}$ can be expressed as

$$F_{2,2} = \int_0^\infty \int_0^{\frac{\rho_0}{a_1 - a_2\rho_0}} \left[1 - \Pr\left(\frac{a_1 X}{a_2 X + a_3 t + 1} \geq \rho_2\right) \times \Pr\left(\frac{a_3 Y}{a_3 z + 1} \geq \rho_2\right)\right] f_Z(z) dz f_T(t) dt. \quad (95)$$

By making using the CCDFs of $X$ and $Y$, $F_{2,2}$ can be rewritten as

$$F_{2,2} = \int_0^\infty \int_0^{\frac{\rho_0}{a_1 - a_2\rho_0}} f_Z(z) dz f_T(t) dt - e^{-\frac{\rho_2}{(a_1 - a_2\rho_2)\lambda_{\mathsf{SN}}} - \frac{\rho_2}{a_3\lambda_{\mathsf{NF}}}} \frac{(K+1)e^{-K}}{\lambda_{\mathsf{SF}}\lambda_{\mathsf{si}}} \times \int_0^\infty \int_0^{\frac{\rho_0}{a_1 - a_2\rho_0}} e^{-\frac{a_3\rho_2 t}{(a_1 - a_2\rho_2)\lambda_{\mathsf{SN}}} - \frac{(K+1)t}{\lambda_{\mathsf{si}}}} \times e^{-\frac{\rho_2 z}{\lambda_{\mathsf{NF}}} - \frac{z}{\lambda_{\mathsf{SF}}}} I_0\left(2\sqrt{\frac{K(K+1)t}{\lambda_{\mathsf{si}}}}\right) dz dt, \quad (96)$$

and after some manipulations, $F_{2,2}$ can be obtained as

$$F_{2,2} = \begin{cases} \Psi(\lambda_{\mathsf{SF}}, \mu_0) - \left(1 - e^{-\left(\frac{\rho_2}{\lambda_{\mathsf{NF}}} + \frac{1}{\lambda_{\mathsf{SF}}}\right)\mu_0}\right) \times \Omega(\rho_2)\Xi(\kappa_2, \alpha_2), & \text{if } \rho_0 < \theta, \rho_2 < \theta, \\ \Psi(\lambda_{\mathsf{SF}}, \mu_0), & \text{if } \rho_0 < \theta, \rho_2 \geq \theta, \\ \Lambda(\rho_2, \kappa_2, \alpha_2), & \text{if } \rho_0 \geq \theta, \rho_2 < \theta, \end{cases} \quad (97)$$

and $F_{2,2} = 1$ if $\rho_0 \geq \theta, \rho_2 \geq \theta$.

Let $F_{2,3} = \Pr(R_{\mathsf{SNF}}^{\mathsf{FD}} \geq R_0, R_{\mathsf{SNF}}^{\mathsf{FD}} < R_2)$, which can be rewritten as

$$F_{2,3} = \Pr(\gamma_{\mathsf{SNF}}^{\mathsf{FD}} \geq \rho_0, \gamma_{\mathsf{SNF}}^{\mathsf{FD}} < \rho_2) = \Pr\left(\min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} \geq \rho_0, \min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_2\right), \quad (98)$$

and after some algebraic steps, $F_{2,3}$ can be obtained as

$$F_{2,3} = \begin{cases} \Lambda(\rho_2, \kappa_2, \alpha_2) - \Lambda(\rho_0, \kappa_0, \alpha_0), & \text{if } \rho_0 < \theta, \rho_2 < \theta, \rho_0 < \rho_2, \\ 1 - \Lambda(\rho_0, \kappa_0, \alpha_0), & \text{if } \rho_0 < \theta, \rho_2 \geq \theta, \end{cases} \quad (99)$$

and $F_{2,3} = 0$ if $(\rho_0 < \theta, \rho_2 < \theta, \rho_0 \geq \rho_2)$ or $(\rho_0 \geq \theta, \forall \rho_2)$.

Let $F_{2,4} = \Pr(R_{\mathsf{SNF}}^{\mathsf{FD}} < R_0, R_{\mathsf{SF}}^{\mathsf{nc}} < R_2)$, which can be rewritten as

$$F_{2,4} = \Pr(\gamma_{\mathsf{SNF}}^{\mathsf{FD}} < \rho_0, \gamma_{\mathsf{SF}}^{\mathsf{nc}} < \rho_2) = \Pr\left(\frac{a_1 Z}{a_2 Z + 1} < \rho_2, \min\left\{\frac{a_1 X}{a_2 X + a_3 T + 1}, \frac{a_3 Y}{a_3 Z + 1}\right\} < \rho_0\right), \quad (100)$$

and after some algebraic steps, $F_{2,4}$ can be obtained as

$$F_{2,4} = \begin{cases} \Psi(\lambda_{\mathsf{SF}}, \mu_2) - \left(1 - e^{-\left(\frac{\rho_0}{\lambda_{\mathsf{NF}}} + \frac{1}{\lambda_{\mathsf{SF}}}\right)\mu_2}\right) \times \Omega(\rho_0)\Xi(\kappa_0, \alpha_0), & \text{if } \rho_0 < \theta, \rho_2 < \theta, \\ \Psi(\lambda_{\mathsf{SF}}, \mu_2), & \text{if } \rho_0 < \theta, \rho_2 \geq \theta, \\ \Lambda(\rho_0, \kappa_0, \alpha_0), & \text{if } \rho_0 \geq \theta, \rho_2 < \theta, \end{cases} \quad (101)$$

and $F_{2,4} = 1$ if $\rho_0 \geq \theta, \rho_2 \geq \theta$.

## Appendix C
## Proof of Proposition 8

From (62), $P_{\text{out},\mathsf{F}}^{\mathtt{sIII}}$ can be rewritten as

$$P_{\text{out},\mathsf{F}}^{\mathtt{sIII}} = \Pr\left(\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < \tilde{\rho}_2, \gamma_{\mathsf{SF}}^{\mathtt{HD}} < \tilde{\rho}_2\right) + \Pr\left(\gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} \geq \tilde{\rho}_2, \gamma_{\mathsf{SF}}^{\mathtt{HD}} + \gamma_{\mathsf{NF}}^{\mathtt{HD}} < \tilde{\rho}_2\right) = \Pr\left(\frac{a_1 X}{a_2 X + 1} < \tilde{\rho}_2, \frac{a_1 Z}{a_2 Z + 1} < \tilde{\rho}_2\right) + \Pr\left(\frac{a_1 X}{a_2 X + 1} \geq \tilde{\rho}_2, \frac{a_1 Z}{a_2 Z + 1} + a_3 Y < \tilde{\rho}_2\right). \quad (102)$$

The result in Proposition 8 can be obtained by utilizing the following lemma with some manipulations.

*Lemma 3:* A following probability can be derived as

$$\underbrace{\Pr\left(\frac{a_1 Z}{a_2 Z + 1} + a_3 Y < \rho\right)}_{P_\Upsilon} = \begin{cases} \Upsilon(\mu, \rho), & \text{if } \rho < \theta, \\ 0, & \text{otherwise.} \end{cases} \quad (103)$$

*Proof:* $P_\Upsilon$ in (103) can be rewritten as

$$P_\Upsilon = \int_0^{\frac{\rho}{a_1 - a_2\rho}} \Pr\left(a_3 Y < \rho - \frac{a_1 z}{a_2 z + 1}\right) f_Z(z) dz. \quad (104)$$

As can be observed, if $\rho \geq \theta$, $P_\Upsilon = 0$. For the case $\rho < \theta$, substituting the CDF of $Y$ and the PDF of $Z$ into (104), $P_\Upsilon$ can be expressed as

$$P_\Upsilon = 1 - e^{-\frac{\mu}{\lambda_Z}} - \frac{1}{\lambda_Z} e^{-\frac{\rho}{a_3\lambda_Y}} \underbrace{\int_0^\mu e^{\frac{a_1 z}{a_2 a_3 \lambda_Y z + a_3\lambda_Y} - \frac{z}{\lambda_Z}} dz}_{I_\Upsilon}. \quad (105)$$

By making a change of variable, i.e., $u = a_2 a_3 \lambda_Y z + a_3 \lambda_Y$, $I_\Upsilon$ can be rewritten as

$$I_\Upsilon = \frac{1}{a_2 a_3 \lambda_Y} e^{\frac{a_1}{a_2 a_3 \lambda_Y} + \frac{1}{a_2\lambda_Z}} \int_{a_3\lambda_Y}^{a_2 a_3 \lambda_Y \mu + a_3 \lambda_Y} e^{-\frac{u}{a_2 a_3 \lambda_Y \lambda_Z} - \frac{a_1}{a_2 u}} du. \quad (106)$$

The integral in (106) can be further expressed as

$$I_\Upsilon = \int_{a_3\lambda_Y}^\infty e^{-\frac{u}{a_2 a_3 \lambda_Y \lambda_Z} - \frac{a_1}{a_2 u}} du - \int_{a_2 a_3 \lambda_Y \mu + a_3 \lambda_Y}^\infty e^{-\frac{u}{a_2 a_3 \lambda_Y \lambda_Z} - \frac{a_1}{a_2 u}} du. \quad (107)$$

In order to solve each integral in (107), we utilize the result in our previous work [32], i.e., the integral $\int_{\mu}^{\infty} e^{-\chi x - \frac{\xi}{x}} dx = \Theta(\mu)$ with $\chi = \frac{1}{a_2 a_3 \lambda_{\mathsf{NF}} \lambda_{\mathsf{SF}}}$, $\xi = \frac{a_1}{a_2}$, $I_{\Upsilon}$ can be obtained as

$$I_{\Upsilon} = \frac{1}{a_2 a_3 \lambda_{\mathsf{NF}}} e^{\frac{a_1}{a_2 a_3 \lambda_{\mathsf{NF}}} + \frac{1}{a_2 \lambda_{\mathsf{SF}}}} [\Theta(a_3 \lambda_{\mathsf{NF}}) - \Theta(a_2 a_3 \lambda_{\mathsf{NF}} \mu + a_3 \lambda_Y)]. \quad (108)$$

By substituting (108) into (105), we complete the proof of Lemma 3. ■

## Appendix D
## Proof of Proposition 9

Let $N_{4,1} = \Pr(R^{\mathtt{HD}}_{\mathsf{SNF}} \geq R_0, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$N_{4,1} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} \geq \tilde{\rho}_0, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} < \tilde{\rho}_2) = \Pr\left( \min\left\{ \frac{a_1 X}{a_2 X + 1}, \frac{a_1 Z}{a_2 Z + 1} + a_3 Y \right\} \geq \tilde{\rho}_0, \frac{a_1 X}{a_2 X + 1} < \tilde{\rho}_2 \right), \quad (109)$$

and after some manipulations, $N_{4,1}$ can be obtained as in (110), and $N_{4,1} = 0$ if $(\tilde{\rho}_0, \tilde{\rho}_2 < \theta, \tilde{\rho}_0 \geq \tilde{\rho}_2)$ or $(\tilde{\rho}_0 \geq \theta, \forall \tilde{\rho}_2)$.

Let $N_{4,2} = \Pr(R^{\mathtt{HD}}_{\mathsf{SNF}} \geq R_0, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$N_{4,2} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} \geq \tilde{\rho}_0, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \tilde{\rho}_2, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}} < \tilde{\rho}_1) = \Pr\left( \min\left\{ \frac{a_1 X}{a_2 X + 1}, \frac{a_1 Z}{a_2 Z + 1} + a_3 Y \right\} \geq \tilde{\rho}_0, \frac{a_1 X}{a_2 X + 1} \geq \tilde{\rho}_2, a_2 X < \tilde{\rho}_1 \right), \quad (111)$$

and after some manipulations, $N_{4,2}$ can be obtained as

$$N_{4,2} = [1 - \Upsilon(\tilde{\mu}_0, \tilde{\rho}_0)][\Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_1) - \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_3)], \quad (112)$$

if $\tilde{\mu}_3 < \theta, \tilde{\mu}_1 > \tilde{\mu}_3$, otherwise, $N_{4,2} = 0$, where $\tilde{\mu}_3 = \big(\max\{\tilde{\rho}_0, \tilde{\rho}_2\}\big) / \big(a_1 - a_2 \max\{\tilde{\rho}_0, \tilde{\rho}_2\}\big)$.

Let $N_{4,3} = \Pr(R^{\mathtt{HD}}_{\mathsf{SNF}} < R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$N_{4,3} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} < \tilde{\rho}_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2) = \Pr\left( \min\left\{ \frac{a_1 X}{a_2 X + 1}, \frac{a_1 Z}{a_2 Z + 1} + a_3 Y \right\} < \tilde{\rho}_0, \frac{a_1 X}{a_2 X + 1} < \rho_2 \right), \quad (113)$$

and after some manipulations, $N_{4,3}$ can be obtained as in (114), and $N_{4,3} = 1$ if $\tilde{\rho}_0 \geq \theta, \rho_2 > \theta$.

Let $N_{4,4} = \Pr(R^{\mathtt{HD}}_{\mathsf{SNF}} < R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$N_{4,4} = \Pr(\gamma^{\mathtt{HD}}_{\mathsf{SNF}} < \tilde{\rho}_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1) = \Pr\left( \min\left\{ \frac{a_1 X}{a_2 X + 1}, \frac{a_1 Z}{a_2 Z + 1} + a_3 Y \right\} < \tilde{\rho}_0, \frac{a_1 X}{a_2 X + 1} \geq \rho_2, a_2 X < \rho_1 \right), \quad (115)$$

and after some manipulations, $N_{4,4}$ can be obtained as in (116), and $N_{4,4} = 0$ if $(\rho_2 \geq \theta)$ or $(\rho_2 < \theta, \mu_1 \leq \mu_2)$.

Let $N_{4,5} = \Pr(R^{\mathtt{nc}}_{\mathsf{SF}} \geq R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$N_{4,5} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} \geq \rho_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} < \rho_2) = \Pr\left( \frac{a_1 Z}{a_2 Z + 1} \geq \rho_0 \right) \Pr\left( \frac{a_1 X}{a_2 X + 1} < \rho_2 \right), \quad (117)$$

and after some manipulations, $N_{4,5}$ can be obtained as

$$N_{4,5} = \begin{cases} [1 - \Psi(\lambda_{\mathsf{SF}}, \mu_0)]\, \Psi(\lambda_{\mathsf{SN}}, \mu_2), & \text{if } \rho_0 < \theta, \rho_2 < \theta, \\ 1 - \Psi(\lambda_{\mathsf{SF}}, \mu_0), & \text{if } \rho_0 < \theta, \rho_2 \geq \theta, \end{cases} \quad (118)$$

and $N_{4,5} = 0$ if $\rho_0 \geq \theta, \forall \rho_2$.

Let $N_{4,6} = \Pr(R^{\mathtt{nc}}_{\mathsf{SF}} \geq R_0, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$N_{4,6} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} \geq \rho_0, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \rho_2, \gamma^{\mathtt{nc}}_{\mathsf{SN},x_{\mathsf{N}}} < \rho_1) = \Pr\left( \frac{a_1 Z}{a_2 Z + 1} \geq \rho_0 \right) \Pr\left( \frac{a_1 X}{a_2 X + 1} \geq \rho_2, a_2 X < \rho_1 \right), \quad (119)$$

and after some manipulations, $N_{4,6}$ can be obtained as

$$N_{4,6} = [1 - \Psi(\lambda_{\mathsf{SF}}, \mu_0)]\, [\Psi(\lambda_{\mathsf{SN}}, \mu_1) - \Psi(\lambda_{\mathsf{SN}}, \mu_2)], \quad (120)$$

if $\rho_0 < \theta, \rho_2 < \theta, \mu_1 > \mu_2$, otherwise, $N_{4,6} = 0$.

Let $N_{4,7} = \Pr(R^{\mathtt{nc}}_{\mathsf{SF}} < R_0, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} < R_2)$, which can be rewritten as

$$N_{4,7} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} < \rho_0, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} < \tilde{\rho}_2) = \Pr\left( \frac{a_1 Z}{a_2 Z + 1} < \rho_0 \right) \Pr\left( \frac{a_1 X}{a_2 X + 1} < \tilde{\rho}_2 \right), \quad (121)$$

and after some manipulations, $N_{4,7}$ can be obtained as

$$N_{4,7} = \begin{cases} \Psi(\lambda_{\mathsf{SF}}, \mu_0) \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_2), & \text{if } \rho_0 < \theta, \tilde{\rho}_2 < \theta, \\ \Psi(\lambda_{\mathsf{SF}}, \mu_0), & \text{if } \rho_0 < \theta, \tilde{\rho}_2 \geq \theta, \\ \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_2), & \text{if } \rho_0 \geq \theta, \tilde{\rho}_2 < \theta, \end{cases} \quad (122)$$

otherwise, $N_{4,7} = 1$.

Let $N_{4,8} = \Pr(R^{\mathtt{nc}}_{\mathsf{SF}} < R_0, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq R_2, R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}} < R_1)$, which can be rewritten as

$$N_{4,8} = \Pr(\gamma^{\mathtt{nc}}_{\mathsf{SF}} < \rho_0, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}} \geq \tilde{\rho}_2, \gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{N}}} < \tilde{\rho}_1) = \Pr\left( \frac{a_1 Z}{a_2 Z + 1} < \rho_0 \right) \Pr\left( \frac{a_1 X}{a_2 X + 1} \geq \tilde{\rho}_2, a_2 X < \tilde{\rho}_1 \right), \quad (123)$$

and after some manipulations, $N_{4,8}$ can be obtained as

$$N_{4,8} = \begin{cases} \Psi(\lambda_{\mathsf{SF}}, \mu_0) \big[\Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_1) - \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_2)\big], & \text{if } \rho_0 < \theta, \tilde{\rho}_2 < \theta, \tilde{\mu}_1 < \tilde{\mu}_2, \\ \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_1) - \Psi(\lambda_{\mathsf{SN}}, \tilde{\mu}_2), & \text{if } \rho_0 < \theta, \tilde{\rho}_2 < \theta, \tilde{\mu}_1 \geq \tilde{\mu}_2, \end{cases} \quad (124)$$

otherwise, $N_{4,8} = 0$.

$$N_{4,1} = \begin{cases} \left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right]\left[\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2)-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right], & \text{if } \tilde{\rho}_0,\tilde{\rho}_2<\theta,\tilde{\rho}_0<\tilde{\rho}_2, \\ \left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right]\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right], & \text{if } \tilde{\rho}_0<\theta,\tilde{\rho}_2\geq\theta. \end{cases} \tag{110}$$

$$N_{4,3} = \begin{cases} \Psi(\lambda_{\mathsf{SN}},\mu_2)-\left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right]\left[\Psi(\lambda_{\mathsf{SN}},\mu_2)-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right], & \text{if } \tilde{\rho}_0<\theta,\rho_2<\theta,\tilde{\mu}_0<\mu_2 \\ \Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } (\tilde{\rho}_0<\theta,\rho_2<\theta,\tilde{\mu}_0\geq\mu_2) \text{ or } (\tilde{\rho}_0\geq\theta,\rho_2<\theta), \\ 1-\left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right]\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right], & \text{if } \tilde{\rho}_0<\theta,\rho_2\geq\theta. \end{cases} \tag{114}$$

$$N_{4,4} = \begin{cases} \Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\left[\Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2)\right], & \text{if } \tilde{\rho}_0<\theta,\rho_2<\theta,\mu_1>\mu_2,\tilde{\mu}_0<\mu_2 \\ \Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2)-\left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right]\left[\Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right], & \text{if } \tilde{\rho}_0<\theta,\rho_2<\theta,\mu_2\leq\tilde{\mu}_0\leq\mu_1 \\ \Psi(\lambda_{\mathsf{SN}},\mu_1)-\Psi(\lambda_{\mathsf{SN}},\mu_2), & \text{if } (\tilde{\rho}_0<\theta,\rho_2<\theta,\tilde{\mu}_0\geq\mu_1>\mu_2) \\ & \quad \text{or } (\tilde{\rho}_0\geq\theta,\rho_2<\theta,\mu_1>\mu_2). \end{cases} \tag{116}$$

## Appendix E
## Proof of Proposition 10

Let $F_{4,1} = \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} \geq R_0, R_{\mathsf{SF}}^{\mathtt{nc}} < R_2)$, which can be rewritten as

$$\begin{aligned} F_{4,1} &= \Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} \geq \rho_0, \gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_2) \\ &= \Pr\left(\frac{a_1 Z}{a_2 Z+1} \geq \rho_0, \frac{a_1 Z}{a_2 Z+1} < \rho_2\right), \end{aligned} \tag{125}$$

and after some manipulations, $F_{4,1}$ can be obtained as

$$F_{4,1} = \begin{cases} \Psi(\lambda_{\mathsf{SF}},\rho_2) \\ \quad -\Psi(\lambda_{\mathsf{SF}},\rho_0), & \text{if } \rho_0<\theta,\rho_2<\theta,\rho_0<\rho_2, \\ 1-\Psi(\lambda_{\mathsf{SF}},\rho_0), & \text{if } \rho_0<\theta,\rho_2\geq\theta, \end{cases} \tag{126}$$

and $F_{4,1}=0$ if $\rho_0 \geq \theta$ or $(\rho_0<\theta,\rho_2<\theta,\rho_0\geq\rho_2)$.

Let $F_{4,2} = \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < R_2, R_{\mathsf{SF}}^{\mathtt{HD}} < R_2)$, which can be rewritten as

$$\begin{aligned} F_{4,2} &= \Pr\left(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0, \gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < \tilde{\rho}_2, \gamma_{\mathsf{SF}}^{\mathtt{HD}} < \tilde{\rho}_2\right) \\ &= \Pr\left(\frac{a_1 Z}{a_2 Z+1} < \rho_0, \frac{a_1 X}{a_2 X+1} < \tilde{\rho}_2, \frac{a_1 Z}{a_2 Z+1} < \tilde{\rho}_2\right), \end{aligned} \tag{127}$$

and after some manipulations, $F_{4,2}$ can be obtained as

$$F_{4,2} = \begin{cases} \Psi(\lambda_{\mathsf{SF}},\hat{\mu}_{02})\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2), & \text{if } \rho_0<\theta,\tilde{\rho}_2<\theta, \\ \Psi(\lambda_{\mathsf{SF}},\tilde{\mu}_2)\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2), & \text{if } \rho_0\geq\theta,\tilde{\rho}_2<\theta, \\ \Psi(\lambda_{\mathsf{SF}},\mu_0), & \text{if } \rho_0<\theta,\tilde{\rho}_2\geq\theta, \end{cases} \tag{128}$$

and $F_{4,2}=1$ if $\rho_0\geq\theta,\tilde{\rho}_2\geq\theta$, where $\hat{\mu}_{02}=\min\{\mu_0,\tilde{\mu}_2\}$.

Let $F_{4,3} = \Pr(R_{\mathsf{SF}}^{\mathtt{nc}} < R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} \geq R_2, R_{\mathsf{F}}^{\mathtt{mrc}} < R_2)$, which can be expressed as

$$\begin{aligned} F_{4,3} &= \Pr(\gamma_{\mathsf{SF}}^{\mathtt{nc}} < \rho_0, \gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} \geq \tilde{\rho}_2, \gamma_{\mathsf{SF}}^{\mathtt{HD}} + \gamma_{\mathsf{NF}}^{\mathtt{HD}} < \tilde{\rho}_2) \\ &= \Pr\left(\frac{a_1 Z}{a_2 Z+1} < \rho_0, \frac{a_1 X}{a_2 X+1} \geq \tilde{\rho}_2, \right. \\ &\qquad \left. \frac{a_1 Z}{a_2 Z+1} + a_3 Y < \tilde{\rho}_2\right). \end{aligned} \tag{129}$$

We first deal with the following probability $F_{4,3,A} = \Pr\left(\frac{a_1 Z}{a_2 Z+1} < \rho_x, \frac{a_1 Z}{a_2 Z+1} + a_3 Y < \rho_y\right)$. Assuming that $\rho_x,\rho_y<\theta$, conditioning on $Z=z$, we rewrite $F_{4,3,A}$ as

$$F_{4,3,A} = \int_0^{\min\{\mu_x,\mu_y\}} \Pr\left(a_3 Y < \rho_y - \frac{a_1 z}{a_2 z+1}\right) f_Z(z)dz, \tag{130}$$

where $\mu_x = \frac{\rho_x}{a_1-a_2\rho_x}$, $\mu_y = \frac{\rho_x}{a_1-a_2\rho_y}$. At this step, by following the same calculation as done in the proof of Lemma 3, we obtain $F_{4,3,A}$ as $F_{4,3,A} = \Upsilon(\min\{\mu_x,\mu_y\},\rho_y)$. Thus, after some manipulations, $F_{4,3}$ can be obtained as

$$F_{4,3} = \begin{cases} (1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2))\Upsilon(\hat{\mu}_{02},\tilde{\rho}_2), & \text{if } \tilde{\rho}_2<\theta,\rho_0<\theta, \\ (1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2))\Upsilon(\tilde{\mu}_2,\tilde{\rho}_2), & \text{if } \tilde{\rho}_2<\theta,\rho_0\geq\theta, \end{cases} \tag{131}$$

and $F_{4,3}=0$ if $\tilde{\rho}_2\geq\theta$, where $\hat{\mu}_{02}=\min\{\mu_0,\tilde{\mu}_2\}$.

Let $F_{4,4} = \Pr(R_{\mathsf{SNF}}^{\mathtt{HD}} \geq R_0, R_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < R_2, R_{\mathsf{SF}}^{\mathtt{HD}} < R_2)$, which can be rewritten as

$$\begin{aligned} F_{4,4} &= \Pr\left(\gamma_{\mathsf{SNF}}^{\mathtt{HD}} \geq \tilde{\rho}_0, \gamma_{\mathsf{SN},x_{\mathsf{F}}}^{\mathtt{HD}} < \tilde{\rho}_2, \gamma_{\mathsf{SF}}^{\mathtt{HD}} < \tilde{\rho}_2\right) \\ &= \Pr\left(\min\left\{\frac{a_1 X}{a_2 X+1}, \frac{a_1 Z}{a_2 Z+1} + a_3 Y\right\} \geq \tilde{\rho}_0, \right. \\ &\qquad \left. \frac{a_1 X}{a_2 X+1} < \tilde{\rho}_2, \frac{a_1 Z}{a_2 Z+1} < \tilde{\rho}_2\right), \end{aligned} \tag{132}$$

and after some manipulations, $F_{4,4}$ can be obtained as

$$F_{4,4}=\begin{cases}\left[\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2)-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right] & \\ \quad\times\left[\Psi(\lambda_{\mathsf{SF}},\tilde{\mu}_2)-\Upsilon(\tilde{\mu}_{02},\tilde{\rho}_0)\right], & \text{if } \tilde{\rho}_2<\theta,\tilde{\rho}_0<\theta,\\ 1-\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right] & \\ \quad\times\left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right], & \text{if } \tilde{\rho}_2\geq\theta,\end{cases} \tag{133}$$

and $F_{4,4}=0$ if $\tilde{\rho}_2<\theta,\tilde{\rho}_0\geq\theta$, where $\tilde{\mu}_{02}=\min\{\tilde{\mu}_0,\tilde{\mu}_2\}$.

Let $F_{4,5}=\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}\geq R_0,R^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq R_2,R^{\mathtt{mrc}}_{\mathsf{F}}<R_2)$, which can be rewritten as

$$\begin{aligned}F_{4,5}&=\Pr\left(\gamma^{\mathtt{HD}}_{\mathsf{SNF}}\geq\tilde{\rho}_0,\gamma^{\mathtt{HD}}_{\mathsf{SN},x_{\mathsf{F}}}\geq\tilde{\rho}_2,\gamma^{\mathtt{HD}}_{\mathsf{SF}}+\gamma^{\mathtt{HD}}_{\mathsf{NF}}<\tilde{\rho}_2\right)\\&=\Pr\left(\min\left\{\frac{a_1X}{a_2X+1},\frac{a_1Z}{a_2X}+a_3Y\right\}\geq\tilde{\rho}_0,\right.\\&\qquad\left.\frac{a_1Z}{a_2Z+1}\geq\tilde{\rho}_2,\frac{a_1Z}{a_2X}+a_3Y<\tilde{\rho}_2\right),\end{aligned} \tag{134}$$

and after some manipulations, $F_{4,5}$ can be obtained as

$$F_{4,5}=\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_2)\right]\left[\Upsilon(\tilde{\mu}_2,\tilde{\rho}_2)-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right], \tag{135}$$

if $\tilde{\rho}_0<\theta,\tilde{\rho}_2<\theta,\tilde{\rho}_0<\tilde{\rho}_2$, otherwise, $F_{4,5}=0$.

Let $F_{4,6}=\Pr(R^{\mathtt{HD}}_{\mathsf{SNF}}<R_0,R^{\mathtt{nc}}_{\mathsf{SF}}<R_2)$, which can be rewritten as

$$\begin{aligned}F_{4,6}&=\Pr\left(\gamma^{\mathtt{HD}}_{\mathsf{SNF}}<\rho_0,\gamma^{\mathtt{nc}}_{\mathsf{SF}}<\rho_2\right),\\&=\Pr\left(\min\left\{\frac{a_1X}{a_2X+1},\frac{a_1Z}{a_2X}+a_3Y\right\}<\tilde{\rho}_0,\right.\\&\qquad\left.\frac{a_1Z}{a_2Z+1}<\rho_2\right),\end{aligned} \tag{136}$$

and after some manipulations, $F_{4,6}$ can be obtained as

$$F_{4,6}=\begin{cases}\Psi(\lambda_{\mathsf{SF}},\mu_2) & \\ \quad-\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right] & \\ \quad\times\left[\Psi(\lambda_{\mathsf{SF}},\mu_2)-\Upsilon(\check{\mu}_{02},\tilde{\rho}_0)\right], & \text{if } \rho_2<\theta,\tilde{\rho}_0<\theta,\\ 1-\left[1-\Psi(\lambda_{\mathsf{SN}},\tilde{\mu}_0)\right] & \\ \quad\times\left[1-\Upsilon(\tilde{\mu}_0,\tilde{\rho}_0)\right], & \text{if } \rho_2\geq\theta,\tilde{\rho}_0<\theta,\\ \Psi(\lambda_{\mathsf{SF}},\mu_2), & \text{if } \rho_2<\theta,\tilde{\rho}_0\geq\theta,\end{cases} \tag{137}$$

and $F_{4,6}=1$ if $\rho_2\geq\theta,\tilde{\rho}_0\geq\theta$, where $\check{\mu}_{02}=\min\{\tilde{\mu}_0,\mu_2\}$.

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